

Highlights

A Review of Multimodal Sentiment Analysis: Taxonomy, Issues, Challenges, and Future Perspectives

Khalid Anwar, Shreya, Meghna Sharma, Kritika Saanvi

- Presents a comprehensive review of multimodal sentiment analysis, highlighting major issues and challenges and their available solutions.
- Discusses the datasets and evaluation metrics used in multimodal sentiment analysis, providing statistical and comparative analysis.
- Presents a compact framework that effectively utilizes the early fusion approach in multimodal sentiment analysis.
- Identifies future research perspectives, offering valuable insights for researchers in multimodal sentiment analysis.

A Review of Multimodal Sentiment Analysis: Taxonomy, Issues, Challenges, and Future Perspectives

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Abstract

Recent developments in computational intelligence have produced a huge volume of multimodal data across different digital platforms. This data is a great source of contextual, sentimental, and emotional information. Multimodal sentiment analysis (MMSA) is the process of inferring sentiments from multimodal data. MMSA has improved the effectiveness and accuracy of sentiment analysis by integrating heterogeneous modalities. However, there are several issues and challenges in combining multiple modalities, like high complexity, modality fusion, lack of explainability, and temporal synchronization. This paper presents a review of MMSA, discussing data modalities, fusion approaches, issues and challenges. It also presents the statistical analysis and overview of datasets and evaluation metrics used in the reviewed papers. Moreover, it identifies several future research opportunities for the research advancements in MMSA. It is believed that the article will be beneficial for the researchers working in the relevant field.

Keywords: Sentiment Analysis, Multimodal Fusion, Multimodal Sentiment Analysis, Information Retrieval, Fusion Approaches, Issues and Challenges.

1. Introduction

In this era of advancements in computational intelligence, a huge amount of data is being generated over the internet [1]. This huge volume of data is highly informative, having information like people's opinions, sentiments,

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emotions, likes, and dislikes about different services, products, government policies, events, and their relationships with other fellow beings [2]. Sentimental analysis (SA) is the process of analyzing a block of data to infer its sentiments using some statistical, machine learning (ML) or natural language processing (NLP) technique [3]. SA has emerged as an important tool in the field of NLP, enabling organizations to interpret public sentiment and accordingly take further decisions to expand their business. Traditional SA approaches primarily focused on text data employing lexicon-based approaches, which utilize predefined lists of words and phrases associated with specific sentiments [4]. ML models such as Naive Bayes, Support Vector Machines (SVM), and Decision Trees involve training algorithms on labeled datasets to classify sentiments as positive, negative or neutral [5]. ML models have performed sentiment classification tasks better than classic lexicon-based approaches but faced difficulty with contextual understanding and subtle nuances in sentiment expression [6]. On the other hand, rule-based systems rely on a set of heuristics handcrafted by developers based on linguistic rules, which mostly show limited scalability and generalization.

Traditional text-based approaches often overlook critical non-verbal cues, such as tone, facial expressions, and body language [7]. Advances in computational intelligence and deep learning (DL) have paved the way for multimodal sentiment analysis (MMSA) [8]. The Internet is full With the vast amount of user-generated data such as articles, review sites, and social media content. MMSA has emerged as a valuable tool for inferring community feedback and sentiments for businesses and research by analyzing this data [9]. MMSA focuses on inferring sentiment by integrating multiple modalities, including text, audio, and visual data [10]. Fortin et al. [11] have demonstrated that sentiment analysis and emotion recognition are more complex tasks when dealing with multimodal data in comparison to text. This may be due to modality heterogeneity, data imbalance, inconsistent fusion, temporal synchronization or lack of interpretability and explainability [12]. These complexities and challenges present new opportunities for researchers to search for efficient and sustainable solutions for multimodal fusion and MMSA. Stappen [13] introduced a framework for MMSA, which facilitates the fusion of continuous emotion annotations and their transformation into discrete emotion classes. Similarly, Abburi [14] proposed a deep neural network approach for sentiment analysis that effectively integrates multiple modalities, including audio and text.

Furthermore, MMSA offers several open research perspectives like mod-

Table 1: Comparison with the existing review studies.

Reference	Modalities	Fusion Approaches	Framework	Issues and Challenges	Datasets	Evaluation Metrics	Future Perspectives
[15]	–	✓	✓	✓	✓	✓	✓
[16]	✓	–	–	✓	✓	✓	–
[17]	–	✓	✓	✓	✓	–	–
[18]	✓	✓	✓	–	✓	–	✓
[19]	–	✓	–	–	✓	✓	–
[20]	✓	✓	✓	✓	✓	✓	–
[21]	–	✓	–	–	✓	✓	✓
This Paper	✓	✓	✓	✓	✓	✓	✓

eling inter-modality and intra-modality dynamics, designing robust MMSA frameworks for real-world applications with low complexity, and developing interpretable and explainable MMSA models that can identify cross-lingual and multilingual sentiments, as well as sentiment causes [22]. Due to all these advancements and challenges, there is a need for a review to synthesize the existing literature, identify the trends, issues, challenges, and future research perspectives in MMSA. Few research studies have reviewed the MMSA. For example, Lai et al. [15] reviewed MMSA highlighting major datasets and fusion approaches. They also discussed evaluation metrics, few issues and future perspectives. Das et al. [16] discussed different unimodal and multimodal sentiment analysis approaches, some benchmark datasets and evaluation metrics in SA and highlighted the major challenges in SA. They reviewed the resources and evaluation approaches in SA and do not discuss the multimodal fusion their issues and future perspectives. Zhu et al. [17] reviewed the MMSA highlighting and discussing the different fusion methods in detail. They also presented general framework for MMSA, discussed popular datasets and some challenges in MMSA. Gandhi et al.[18] presented a comprehensive review of MMSA, discussing the data modalities, fusion approaches, datasets, applications, challenges. However, they do not do not present the statistical analysis and evaluation metrics. Abdu et al. [19] reviewed the multimodal video sentiment analysis using deep learning. They highlighted benchmark datasets in MMSA, discussed the process and analyzed the performance of some of the state-of-the-art approaches in multimodal video sentiment analysis. Pandey and Vishwakarma [20] comprehensively reviewed SA outlining different data modalities, fusion approaches, machine learning and deep learning approaches for sentiment analysis. They also discussed the benchmark datasets and applications of SA. However, they do not discuss the challenges and futures research perspectives in MMSA.

Zhao et al. [21] reviewed multimodal aspect based SA discussing benchmark datasets, fusion approaches, evaluation metrics and some future perspectives in MMSA. However, they do not focus on the data modalities and issues in MMSA. This paper comprehensively reviews MMSA, highlighting taxonomy, advancements, challenges, and future prospects. It systematically identifies critical challenges, including computational complexity, text-centric pre-trained models, inefficiencies in data fusion, inconsistent data quality, and limited real-time adaptability. It reviews several high-quality papers and summarizes the solutions presented in these papers to overcome these issues. Moreover, it presents a general framework for MMSA. It also discusses the available benchmark datasets and evaluation metrics that are used by researchers to evaluate the MMSA models. Furthermore, it proposes future research perspectives in MMSA. Table 1 highlights the contribution of this paper and compares it with the existing review studies. Figure 1 presents the taxonomy of the paper. The main contributions of this paper can be summarized using following points.

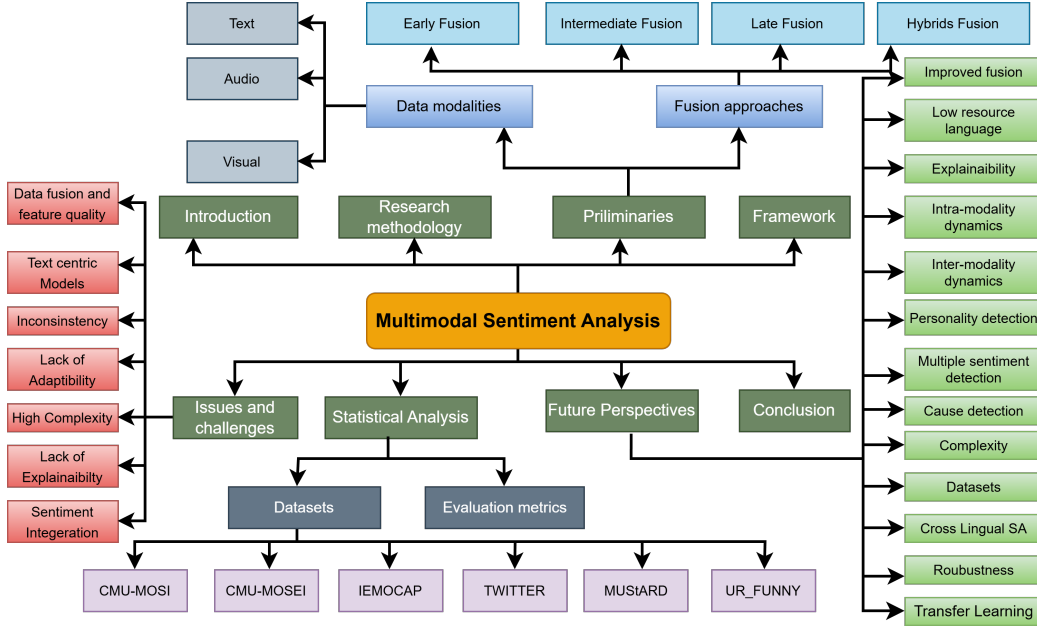


Figure 1: Taxonomy of the paper.

- Presents a comprehensive review of MMSA, highlighting major issues and challenges and their available solutions.

- Discusses the datasets and evaluation metrics used in the reviewed papers, providing statistical and comparative analysis.
- Presents a compact framework that effectively utilizes the early fusion approach in MMSA.
- Identifies future research perspectives, offering valuable insights for researchers in MMSA.

The remaining sections of this paper are organized as follows: Section 2 outlines the research methodology, Section 3 discusses the preliminaries, including the data modalities and fusion approaches. Section 4 presents a general framework for MMSA. Section 5 discusses the issues and challenges in MMSA. Section 6 presents statistical and comparative analysis, highlighting benchmark datasets and evaluation metrics. Section 7 proposes future research perspectives in MMSA. Section 8 is the conclusion.

2. Research methodology

This section presents the methodology adopted to conduct and write this review. It will discuss the paper selection process, including the publication venues, search strings, inclusion, and exclusion criteria.

2.1. Article selection procedure

The articles included in this review were selected through a systematic search strategy. Several digital libraries and databases, including IEEE Xplore, Elsevier, Springer, ACM Digital Library, and arXiv, were searched using Keywords such as "multimodal fusion", "sentiment analysis", "fusion approaches", and "multimodal datasets". Papers published between 2015 and 2025 were included, with a focus on peer-reviewed journals and high-quality conference proceedings. In the initial screening, papers were selected based on the review of the title and abstract, followed by a full-text review. The Inclusion criteria required that the work addressed multimodal sentiment analysis using at least two modalities and provided experimental evaluation. Papers limited to unimodal sentiment analysis or without empirical validation were excluded from the review process.

3. Preliminaries

MMSA approaches infer sentiments from multiple data modalities like text, audio, and video to recognize the expressed emotion using intelligent computing techniques. MMSA utilizes multiple data modalities to capture human emotion to a greater extent to enhance sentiment understanding. MMSA can operate in unimodal, bimodal, and trimodal configurations. The subsequent subsections presents the fundamentals of MMSA, discussing different data modalities and fusion approaches.

3.1. Data modalities

MMSA use multiple data modalities to infer sentiments. It involves integrating text, audio, and visual data, like images and videos, to provide a more comprehensive understanding of sentiments [23]. It adds richness to the data and leads to better accuracy and insights than unimodal analysis, which depends on just one type of data [24]. An overview of different data modalities is presented in the subsequent subsections.

3.1.1. Texts

Text is among the most widely utilized data modalities in SA due to its capacity to convey opinions, emotions, and subjective expressions. This unstructured data is commonly found in online reviews, social media posts, blogs, and news articles, among other web-based platforms [25]. To extract meaningful information, text data is preprocessed and transformed into a structured format. Subsequently, various NLP techniques, including dictionary-based, lexicon-based, ML methods such as SVM and decision trees, as well as deep learning (DL) approaches like recurrent neural networks (RNN), long short-term memory (LSTM), and BERT, are employed to infer sentiments [26]. Challenges in text-based SA include difficulties in addressing sarcasm, slang, ambiguous language, and multilingual content [27].

3.1.2. Audio

Audio data encompasses the rich landscape of speech and sound, capturing the nuanced emotions conveyed through vocal expression. It focuses on acoustic features like tone, pitch, volume, and prosodic variations. Sentiment embedded within spoken language can be effectively analyzed through various speech processing and DL techniques [28]. Commonly utilized datasets in this domain comprise speech recordings sourced from diverse contexts

such as podcasts, interviews, customer service interactions, and cinematic dialogues. However, analyzing audio data presents several challenges, particularly the confounding effects of background noise, inter-speaker variability, and cross-cultural differences in vocal expression patterns. These complexities underscore the necessity of developing robust and generalizable models capable of reliable sentiment detection [29].

3.1.3. *Visuals*

The visual data comprises of facial expressions, body language and visual attention. They are rich source of sentiments. They deliver a much richer perspective on sentiment than modalities such as text and speech due their ability to represent the features ignored by and text and speech and provide more accurate predictions of emotion states [30].

Vision-based sentiment analysis pays more attention to facial expressions and body language, adding non-verbal emotional cues to SA. Facial expressions can convey feelings of contentment, subtle anger, and sometimes sarcasm [31]. Although image processing has become a prominent field of research in computer vision, researchers face the issue of the quality of images and differences in the cultural expression of emotions. Body Language is also involved in the process of analyzing sentiment. Because through body language, there is a manifestation of emotions or attitude, though not voiced in words or texts or any speeches. Such body language features as crossed arms, open gestures, slumped posture, or leaning forward can be an indication of states of sentiment like defensiveness, openness, confidence, or nervousness. The analysis of body movements together with speech tone and textual content can contextualize the overall emotional state of the individual.

3.2. *Multimodal data fusion approaches*

Multimodal data fusion (MMDF) refers to the process of combining information from various modalities (e.g., text, audio, images) to enhance sentiment analysis accuracy. Multimodal fusion techniques are critical in MMSA for effectively integrating diverse data sources [32]. However, integrating multimodal features is challenging due to irrelevant information interference and incomplete fusion [33]. To overcome these challenges, researchers proposed multiple fusion approaches. These fusion approaches can be categorized as early fusion, late fusion, intermediate fusion, and hybrid fusion [34]. A brief overview of these MMDF approaches is presented in the subsequent sub-

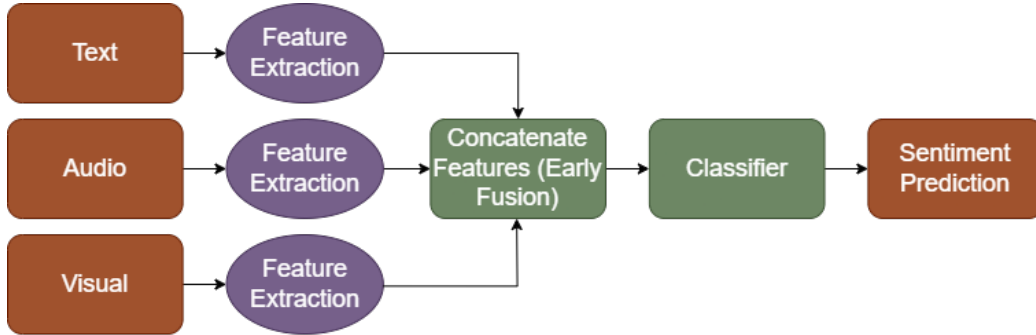


Figure 2: Early Fusion

sections. Table 2 discusses the strengths and weaknesses of different fusion approaches.

3.2.1. Early fusion

In early fusion, data from different modalities are combined at the feature level, before the actual model training, as shown in Figure 2. Raw or low-level features (e.g., pixel values from images, frequency components from audio, or word embeddings from text) are extracted, normalized, and then integrated into a single feature representation [35]. The early fusion approach utilizes the power of feature-level information integration and is simple and easy to implement [36]. However, it produces large input vectors having redundant information [37]. Moreover, it fails to handle intra-modality dynamics like timing mismatch, which may lead to poor performance [38].

3.2.2. Late fusion

Late fusion combines data at the decision level, meaning each modality is processed separately through its model or classifier. The outputs are then merged to make a final decision [39]. Each modality is treated independently, and the individual models can reach their conclusions as shown in Figure 3. These individual outputs are fused, often using techniques like majority voting, weighted averaging, or ensemble learning [40][41]. Late fusion approaches are more flexible and scalable to different modalities, and their modular approach allows for modality-specific classifiers. They are easier to manage in conflicting modalities [42]. The aggregation methods, like averaging and voting, may fail to capture inter-modality dynamics. Moreover, late approaches are time-consuming due to weight optimization in multiple classifiers [43].

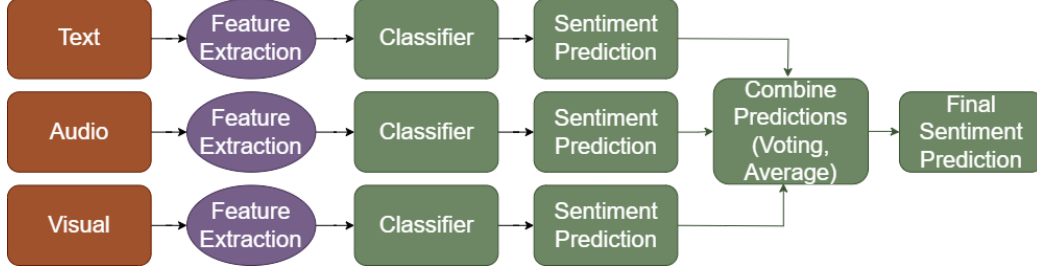


Figure 3: Late Fusion

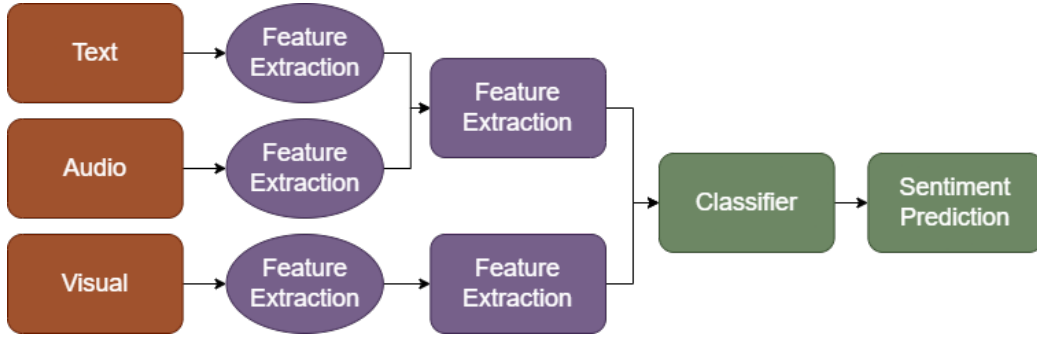


Figure 4: Intermediate Fusion

3.2.3. Intermediate fusion

Intermediate fusion is mainly realized through the neural network, sharing the middle layer of the shared network during the fusion process [44]. Intermediate fusion is a middle-ground approach where features from different modalities are fused after initial processing but before the final decision. This fusion usually occurs at a middle layer of the model, where each modality has its own initial processing or encoding stage, as shown in Figure 4. Each modality goes through its feature extraction process (e.g., an embedding layer or a convolutional layer for images, an RNN layer for text) [45]. The intermediate features are then combined using methods like concatenation or cross-attention to form a single feature representation, which is further processed for the final prediction [46][47]. The intermediate fusion approaches achieve state-of-the-art results by combining intra-modality and inter-modality interaction. However, the performance may degrade if any modality has incomplete or noisy data [48].

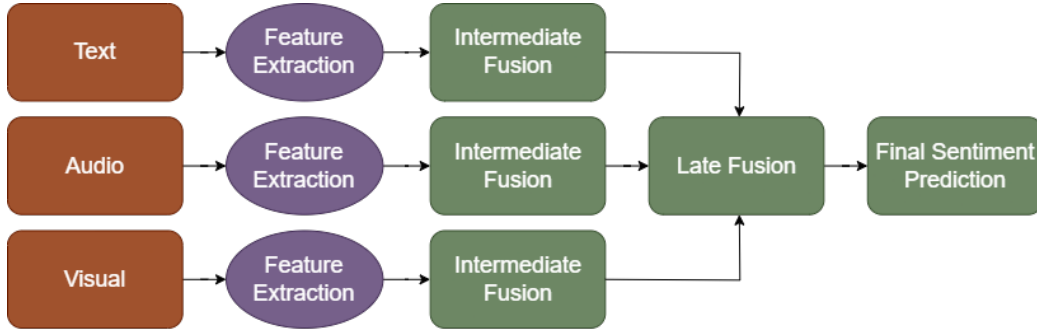


Figure 5: Hybrid fusion

3.2.4. Hybrid fusion

Hybrid fusion combines aspects of early, intermediate, and late fusion approaches within the same model, as shown in Figure 5. This fusion approach allows data from different modalities to be fused at multiple levels within the model to capture both low-level and high-level interactions [49]. Some information from the modalities may be fused at the feature level, while other information is fused at intermediate layers or at the decision level, allowing flexibility to extract information at different depths [50]. The hybrid fusion approaches combine the advantages of early and late fusion to model intra-model and cross-model dynamics[51]. They can better handle inter-modality interactions. However, these approaches are complex and may have information redundancy issues [52].

Table 2: Strengths and weaknesses of different fusion approaches.

Fusion	Strength	Weakness
Early fusion [53, 54, 55]	• Simple and computationally efficient. • Enables richer representation through modality combination using feature-level integration. • Performs well when modalities are fully aligned. • Captures the cross-modal correlations at the feature level.	• Prone to overfitting. • Ineffective in modeling intra-modality dynamics. • The complexity increases with the increase in the number of modalities. • Misaligned modalities and imprecise time synchronization hinder learning performance.
Late fusion [56, 57, 58]	• Enhances robustness to missing or noisy modalities. • Suitable for heterogeneous and asynchronous modalities. • Modality-specific modeling and optimization facilitate the interpretability of individual modality contributions.	• Fails to capture cross-modal dynamics. • May lead to sub-optimal integration in tasks requiring strong cross-modal dependencies. • Fusion often relies on heuristic-based decision combination, which may not generalize well.
Intermediate fusion [40, 45, 59]	• Balances between early and late fusion in terms of feature integration and model complexity. • Merges modality-specific paths at intermediate network layers for better correlation modeling. • Effective for structured and deep fusion architectures (e.g., multimodal BERT).	• More complex to implement compared to early or late fusion. • Requires careful design to optimize information flow between modalities. • Performance degrades if some multimodal data is incomplete or noisy.

Hybrid fusion [56, 57, 51]	• Combines the advantages of multiple fusion paradigms. • Offers flexibility to capture both feature-level and decision-level interactions. • Highly adaptable to real-world multimodal applications and hierarchical decision-making.	• May suffer from data or information redundancy issues. • Harder to interpret and analyze due to multi-level fusion logic. • Implementation can be complex and may require advanced computing resources.
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4. General framework for multimodal sentiment analysis

This section presents a general framework for multimodal sentiment analysis using early fusion. The framework consists of multiple phases as shown in Figure 6 . Different phases of the framework are data input, data pre-processing, feature extraction, feature fusion, and sentiment classification. A brief overview of the various phases is presented below.

The first phase of the framework is the data input phase. In this phase, the data is collected and fed to the model. The input data can be in different modalities: text, visual (image and video), and audio. The input data may be unstructured and have various types of noise. The noisy and unstructured data must be preprocessed and converted to a well-structured format for enhanced model performance [60]. The data pre-processing is performed in the second phase of the general framework. There are different data pre-processing techniques for different data modalities like tokenization, lemmatization, and stop word removal for text data pre-processing, noise and silence reduction for audio data, noise reduction and grayscale conversion for visual data [61]. The pre-processing phase converts the data into a well-structured format that is free of noise. Subsequently, the clean data are passed to the third phase to select and extract important features. The feature extraction approaches for different data modalities are different. For example, bag-of-words and tf-idf are used for feature extraction in textual data. In addition to these classical methods, recent transformer-based language models such as BERT, RoBERTa, and XLNet have been widely used in textual feature extraction as well. These models provide contextualized embeddings, which capture semantic information and therefore are best suited to multimodal

applications in which text accompanies audio and visual modalities. BERT embeddings, unlike TF-IDF, are incidentally frequency-based and sparse, providing dense and contextualized embeddings, which allow sentiment inference in multi-faceted social media and conversational applications. spectrogram and mel-frequency cepstral coefficients (MFCCs) are used for feature extraction in audio data. There are several commonly-used open-source tools available for feature extraction in audio data. OpenSMILE can be utilized to extract default prosody, spectral, and voice quality features; COVAREP provides higher-level speech analysis features like glottal source and prosody descriptors; and LibROSA, being in Python, can frequently be seen to extract spectral and time-series features. These publicly available, lightweight tools constitute the foundation of most MMSA pipelines. Optical flow and facial landmarks are some of the frequently used feature extraction techniques in visual data [62]. Beyond classical hand-crafted visual attributes like HOG and LBP, recent deep learning models based on convolution and transformer architecture have become mainstream for extracting visual features. CNNs (e.g., VGG, Inception, ResNet) extract hierarchical spatial information in face expressions, hand gestures, and scenes, whereas recent transformer-based models like TimeSformer and Vision Transformer (ViT) further this ability to model spatial and temporal dependencies. These models extract rich, contextual visual embeddings, which prove very useful for multimodal processing, particularly in combination with textual and acoustic features. In contrast to hand-crafted descriptors, whose robustness and discriminability are limited, CNN- and ViT-based embeddings offer richer, more complete representations, and better support sentiment inference under dynamic and unrestricted settings.

Prior to feature fusion, modalities undergo temporal synchronisation to guarantee that text, audio, and video features are aligned in time. Text–speech synchronisation at the word level is usually carried out through forced alignment methods (i.e., Montreal Forced Aligner) that align spoken words to specific time stamps. Video frames are sampled at a fixed sampling frequency (25–30 fps) and synchronized to the time intervals of speech segments being aligned. Acoustic features (spectrograms or MFCCs) are estimated over short overlapping segments (20–40 ms) and pooled to match utterance or word boundaries. Interpolation or downsampling is finally carried out to guarantee all modalities have one common temporal index before feature fusion. This step helps ensure cross-modal correlations are maintained and boosts early fusion-based sentiment classification accuracy.

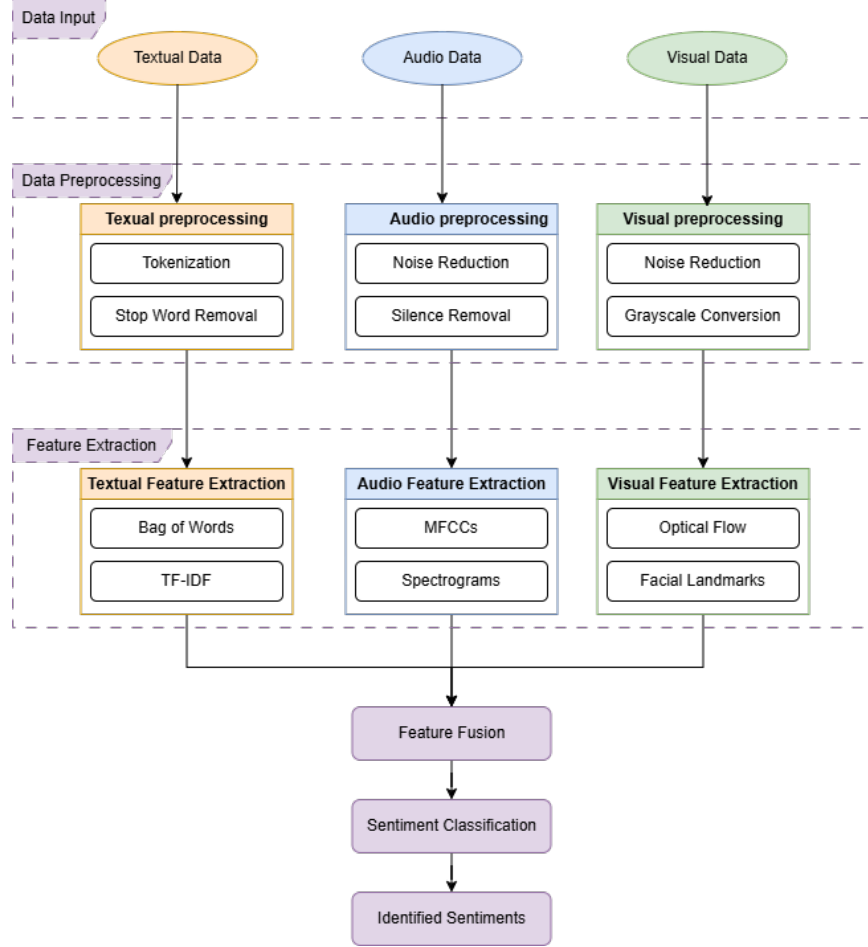


Figure 6: General framework for multi-modal sentiment analysis.

The next two steps in MMSA are feature fusion and sentiment classification. These two steps are interchanged according to the fusion strategy adopted. This framework presents an early fusion approach, where fusion occurs before the application of the machine learning classifier [63].

To eliminate domination of high-dimensional modalities (e.g., visual embeddings) over low-dimensional modalities (e.g., prosodic audio features) in early fusion, feature normalization and dimension balancing occur prior to fusion. Firstly, feature scaling methods such as z-score normalization or min-max scaling normalize all modality features to an identical numeric range [35]. Secondly, dimensionality reduction methods such as Principal

Component Analysis (PCA), linear projections, or neural-network-based bottleneck layers reduce high-dimensional visual embeddings to lower-dimensional latent spaces [38]. Accordingly, low-dimensional features can be upscaled or projected to an identical latent space by dense layers to ensure dimension compatibility. Furthermore, modality weighing and attention-based schemes can be incorporated to control adaptively each modality’s contribution in fusion [40, 57]. Learnable modality-specific weights allow the model to attend to most-informative modality features and inhibit over-reliance on higher-dimensional features. Lastly, regularization schemes (e.g., dropout, L2 regularization) act to suppress overfitting, potentially due to a high-dimensional fused representation [37]. Individually, these steps allow for balanced, robust multimodal feature representation and sentiment classification [64].

To promote trust and transparency, post-hoc explainability techniques, which provide quantitative measures of each modality’s contribution to the final prediction, may be used. Feature attribution techniques, such as SHAP and LIME, can be applied to the fused feature representation to approximate local importance scores of text, audio, and visual inputs [65, 66]. Additionally, modality saliency maps can be extracted to highlight influential regions of video frames or audio spectrograms, and attention-weight visualization in early- or late-fusion models can be utilized to comprehend upon which modality the model paid attention to reach a prediction in the input [67]. As a whole, these techniques provide for human-interpretable explanations, facilitating model decision verification and potential modality biases diagnostics.

5. Issues and challenges in multimodal sentiment analysis

MMSA is an emerging area of research that integrates textual, audio and visual data for SA. It make use of non-verbal clues like tone, gestures, and facial expressions in addition to text data for a thorough analysis of sentiments. The intricacy of dealing with various and frequently fragmented data modalities presents the field with several challenges [68]. Intelligent fusion techniques are crucial for integrating multimodal data to resolve inherent inconsistencies and ensure the representation of meaningful information [69]. Moreover, scalability and lack of interpretability are major issues. Therefore, there is a need for a careful balance between increasing model transparency and prediction accuracy. Many models are uninterpretable, which reduces their credibility and prevents wider adoption [70]. The cultural and contex-

tual variations in different languages and their syntax raise issues that need to be addressed using intelligent computing approaches. This section summarizes the major issues and challenges identified in the reviewed MMSA literature. It also discusses the solutions proposed by various researchers to overcome these issues.

5.1. Data fusion and feature quality

The fusion of various data modalities poses a serious challenge due to variation in feature qualities [71]. Giving equal weightage to different features may lead to repetitive information and decreased prediction accuracy. Achieving the best feature quality when combining multimodal data is challenging, especially when modalities have distinct properties. Researchers developed multiple approaches for efficient data fusion and improved performance of MMSA. For instance, Multimodal Translation for Sentiment Analysis (MTSA) [53] reduces discrepancies in visual and audio data by transforming them into text-based representations. Deep Emotional Arousal Network (DEAN) [56] reduces redundancy by dynamically adjusting modality contributions using a multimodal gating block. Multimodal Adaptation Gate (MAG+) improves the quality of the fused multimodal representations by reinforcing each modality via cross-modal attention before fusion [72]. Table 3 presents the summary of the representative approaches for data fusion and feature quality improvement in MMSA, showing contribution, datasets, modalities and approaches.

Table 3: Representative approaches for data fusion and feature quality in MMSA

Ref.	Contribution	Dataset	Modalities	Approaches
[53]	Reduces Discrepancy by translating visual and audio data into textual representation and multimodal integration using BERT.	CMU-MOSI, CMU-MOSEI	Text, Audio, Video	MTSA, BERT
[56]	Improves MMSA performance by processing different modalities independently and reducing redundancy.	CMU-MOSI, CMU-MOSEI, IEMOCAP	Text, Image, Audio	CNN and DEAN

Ref.	Contribution	Dataset	Modalities	Approaches
[57]	Resolves modality alignment problem by using multi head attention mechanism.	CH-SIMS, CMU-MOSI, MOSEI, IEMOCAP	Video, Audio	Multi-head Attention
[73]	Reduces processing costs by projecting features into a four-dimensional space. Integrates CNN and fuzzy logic for better sentiment prediction.	TripAdvisor, MOUD, Amigos	Text, Video, ECG Signals	Fuzzy logic, CNN, RNN
[74]	Proposed Attention-based Causality-Aware Fusion (AtCAF) for reducing spurious correlations and improving robustness.	CMU-MOSI, CMU-MOSEI, UR-FUNNY	Text, Video, Audio	Counterfactual Cross-modal Attention (CCoAt)
[75]	Proposed SKEAFN, a fusion network incorporating external sentiment knowledge with attention-based text-guided interaction	CMU-MOSI, CMU-MOSEI, Twitter2019	Text, Visual, Acoustic	Sentiment Knowledge Enhanced Attention Fusion Network (SKEAFN)
[76]	Proposed Atlantis, a trident-shaped framework with textual-vision alignment, for enhancing feature quality, and improving integration of textual and visual data.	Twitter-2015, Twitter-2017	Text, Image	Aesthetic-oriented Multiple Granularities Fusion Network (Atlantis)

5.2. Text-centric models

Textual input is frequently given priority in pre-trained MMSA models, which restricts their ability to include additional modalities and adjust to various situations. Due to due to text density in MMSA models, there is often more abstract and explicit text information than in audio or visual streams, and text is the most dominant modality in common datasets like CMU-MOSI and CMU-MOSEI. Nevertheless, text over-reliance often suppresses the contribution of non-text modalities, resulting in partial sentiment

understanding. To overcome this imbalance, data and model-centric countermeasures have been investigated. At the data level, attempts have been made to compile better-balanced corpora in which visual or acoustic indicators have an increased role, for instance, in CH-SIMS and MUsTARD. At the model level, models such as multimodal translation for sentiment analysis (MTSA) [53] transform visual and acoustic inputs to text-based inputs in order to minimize disparities and retain non-text indicators. Similarly, sentiment-aware multimodal pre-training (SMP) [77] combines sentiment information from textual and visual information to prevent text-only overfitting. Contrastive learning modules, such as cross-modal contrastive learning (CCL) [78] assist in bringing text in sync with other modalities, resulting in more balanced multimodal representations. The text-centric hierarchical fusion network (TCHFNet) [59] uses contrastive learning and cross-modal interactions to match text with non-text modalities. These attempts confirm an increased realization that minimizing text dominance can increase robustness and provide more complete sentiment understanding. The sentiment contrastive learning (SCL) [79] framework integrates the emotional context into the representation learning process, improving the overall model’s performance and comprehension of sentiments across modalities. Table 4 presents a summary of the representative MMSA approaches developed to mitigate the text-centric models.

Table 4: Representative approaches to mitigate text-centric MMSA.

Ref.	Contribution	Dataset	Modalities	DL Technique
[59]	Introduces TCHFN using text-centric contrastive learning with hierarchical fusion to reduce redundancy and enhance the role of text.	CMU-MOSI, CMU-MOSEI, CH-SIMS	Text, Audio, Video	Cross-modal Reinforced Transformer, Knowledge Distillation, Contrastive Learning
[77]	A sentiment-aware multimodal pre-training (SMP) framework for multimodal sentiment analysis.	VSO, MVSA, Twitter15, Twitter17	Text, Image	Pre-training, Transfer learning, CCL
[78]	Presented a first-of-its-kind Persian multimodal dataset and developed a context-aware multimodal fusion framework using early and late fusion techniques for sentiment analysis in Persian videos.	Custom Persian text-image dataset	Text, Audio, Visual	BiLSTM, 3D-CNN, MLP
[79]	Proposed Sentimental Contrastive Learning (SCL) framework, integrating sentimental information using instance-level and cluster-level contrastive learning.	NYT Gigaword Corpus	Text	Sentimental Contrastive Learning (SCL)

5.3. Inconsistent data quality

Inconsistent data quality across modalities, such as noisy audio signals and variable feature reliability, provides difficulties for MMSA and can impair the precision of examining intricate social media information[80][81]. The Text-Image Bidirectional Emotion Recognition Transformer (TIBERT), which aims to capture both within-modality and cross-modality interactions, was proposed by Yu Wei et al.[82] to address these problems. In order to improve sentiment recognition and capture minute emotional nuances, the model integrates text and visual data using sophisticated fusion algorithms. Its potential for industrial applications is highlighted by experimental results showing a 4% accuracy gain over single-modality techniques on both public and proprietary datasets. The Deep Multimodal Attentive Fusion (DMAF) model was presented by Huang Zhan.[54] in order to reduce data inconsistencies. To record interactions between textual and visual modes, it employs a hybrid fusion architecture. The model extracts important sentiment features using unimodal attention approaches and integrates features using a multimodal attention model. The final emotion forecasts are then generated by late fusion. Across several datasets, the DMAF model continuously outperforms current techniques, exhibiting greater sentiment detec-

tion through a deeper comprehension of image-text interactions. Similarly, TriDiRA [83][84], a triple disentangled representation learning approach, resolves Inconsistent Data Quality by preventing models from being perturbed by irrelevant and conflicting information across modalities. By enhancing the disentanglement of modality-invariant, modality-specific, and label-specific representations, TriDiRA improves the consistency and quality of multimodal data, thereby enhancing sentiment analysis performance. Additionally, the model effectively handles sparse and tenuous connections between sarcastic features within and across modalities, ensuring more reliable predictions. A growing trend to resolve inconsistent data quality is missing modality imputation, which generates unavailable modalities from observed ones prior to fusion. Approaches to modality hallucination involve employing deep generative models (i.e., conditional GANs, VAEs) to synthesize synthetic visual or audio embeddings given text features. Lin and Hu[85] introduce MissModal, a method that aligns the representations of complete and incomplete data in MMSA. It leverages geometric contrastive loss and sentiment semantic loss to enhance robustness against missing modalities, all without requiring changes to the fusion stage. The SAR model (Scanning, Attention, and Reasoning) proposed by Liu Li et al.[86] uses a three-step process to extract emotions from multimodal social media content. Heterogeneous Reasoning uses graph neural networks to capture complex interactions inside social networks, Perceptual Scanning performs initial evaluations of text and images, and Intensive Attention uses attention mechanisms for cross-modal feature connections. The SAR model addresses the challenges of multimodal social media analysis by integrating complementary factors and modality interrelations to guarantee accurate and trustworthy sentiment analysis [87]. However, their performance may be negatively impacted by an excessive dependence on noise effects and synthetic features. Therefore, imputation techniques should be developed that effectively manage uncertainty and boost resilience without compromising reliability. Table 5 summarizes the approaches proposed to address inconsistent data quality in MMSA.

Table 5: Representative approaches to overcome inconsistent data quality

Ref.	Contribution	Dataset	Modalities	DL Technique
[82]	Developed feature-guided multimodal sentiment analysis model with attention mechanism	Weibo, MVSA-Single	Text, Image	Attention Mechanism
[54]	Proposed Deep Multimodal Attentive Fusion (DMAF), which integrates intermediate and late fusion for better multimodal sentiment classification by capturing complementary information.	Getty Image, Twitter, Flickr	Text, Images	Multimodal Attention Model (VAM, SAM, MAM, DMAF)
[86]	Proposed the Scanning, Attention, and Reasoning (SAR) model that uses a multi-step reasoning process with graph neural networks.	Flickr, Getty Images, Twitter	Image, Text, Social Networks	Graph Neural Network (GCN), Memory Network
[83]	Proposed TriDiRA, a triple disentanglement model that separates modality-invariant, effective modality-specific, and ineffective modality-specific representations to enhance multimodal affective analysis.	CMU-MOSI, CMU-MOSEI, MELD, UR-FUNNY	Text, Audio, Video	Triple Disentanglement with Attention
[80]	Proposed SAHFN, a hierarchical fusion network integrating sentiment-aware attention and inter-modal transformers for improved sarcasm detection performance.	Twitter Multimodal Sarcasm Dataset	Text, Image, Attribute	Sentiment-aware Hierarchical Fusion Network (SAHFN)
[84]	Proposed TriDiRA, a triple disentanglement approach separating modality-invariant, effective modality-specific, and ineffective modality-specific representations.	CMU-MOSI, CMU-MOSEI, UR-FUNNY, MELD	Text, Visual, Audio	Triple Disentanglement Representation Learning (TriDiRA)
[81]	Proposed MIECF: Multi-faceted Information Extraction and Cross-Mixture Fusion framework for MABSA.	Twitter-2015, Twitter-2017	Text, Image	Multi-faceted Information Extraction (MIE) and Cross-Mixture Fusion (CF)
[87]	Proposed a stochastic multimodal transformer leveraging audio, facial expressions, and text for PTSD detection.	eDAIC	Audio, Video, Text	Multimodal Fusion, Temporal Attention, LWTA Layers

5.4. Lack of real-time adaptability and temporal alignment

Current sentiment analysis techniques’ incapacity to adapt to real-time circumstances restricts their application in contexts requiring accurate and timely responses. In real-world MMSA applications, missing modalities are a frequent issue, particularly in contexts like social media. The multitask framework, which can handle missing modalities by training with text-only and image-only examples [11], is one strategy to improve resilience and generalization. The Unified Multimodal Framework (UniMF) offers a robust solution to these problems by using the Multimodal Generation Transformer (MGT) to generate missing modalities and the Multimodal Understanding Transformer (MUT) to handle unaligned sequences, ensuring accurate and efficient sentiment analysis even in the face of incomplete data [88]. The multimodal attention technique based on gated emoji was proposed by Chauhan

Singh [89]. Its primary objective is to detect sarcasm; sentiment and emotion analysis are secondary goals. This method combines inputs like text, tone, and emojis to detect minute emotional cues like sarcasm and subtle tone shifts. Its attention technique dynamically selects and highlights the most relevant features, enabling accurate sentiment detection even in settings that vary rapidly and entail frequent variations in emotional states. In an extra effort to enhance real-time flexibility, Guo et al. [55] suggested the Text Enhanced Transformer Fusion Network (TETFN). This approach reduces the challenges posed by unpredictable visual and auditory inputs in dynamic contexts by use text as the main medium. TETFN uses text-driven cross-modal mappings to generate comprehensive sentiment representations, and its use of the Vision-Transformer (ViT) ensures the effective gathering of both global and detailed visual data. TETFN’s multi-head attention mechanism prioritizes emotionally significant material, ensuring outstanding relevance and interpretability. These qualities make TETFN a strong and flexible framework that has proven its ability to do sentiment analysis in real time under a range of conditions and generated excellent results on benchmark datasets. Lower sentiment identification accuracy and challenges integrating data streams are often caused by MMSA’s incapacity to correctly align different types of data. Hyperbolic Learning for multimodal sentiment and emotion recognition is proposed by Arano et al. [90]. It captures hierarchical and temporal correlations across multimodal input by utilizing the hyperbolic geometry of neural networks. This method enables the model to more precisely represent complex data interactions by mixing Euclidean and hyperbolic layers. Time-sensitive data streams from textual, visual, and audio modalities are better synchronized by the framework thanks to these layers. This hybrid architecture’s remarkable capacity to handle and synchronize inputs across time significantly improves the alignment and integration of multimodal data streams. By effectively resolving temporal alignment concerns, this method increases overall sentiment classification accuracy, according to empirical results from benchmark datasets like CMU-MOSEI and CMU-MOSI. Current MMSA models use contrastive and causal learning objectives to directly address incongruence between modalities. For example, DEAN uses contrastive loss to separate incongruent modality pair embeddings and align semantically congruent cross-modal embeddings [56]. In a similar vein, AtCAF [74] uses a causal intervention paradigm that separates spurious correlations resulting from opposing facial expressions and text from sentiment-relevant causal features. When combined, these methods demon-

strate that modality-incongruent robustness—a crucial component for practical implementation—is greatly enhanced by contrastive and causal learning. Table 6 presents a summary of the discussed approaches.

Table 6: Techniques for real-time adaptability and temporal alignment

Ref.	Contribution	Dataset	Modalities	DL Technique
[89]	Extended MUSTARD to SEEmoji MUSTARD by annotating additional sarcasm, emotion, and emoji data.	SEEmoji MUSTARD	Text, Audio, Visual	Bi-GRU
[55]	Proposed Text Enhanced Transformer Fusion Network (TETFN) to enhance the textual modality’s contribution to fusion.	CMU-MOSI, CMU-MOSEI	text, video, audio	Vision Transformer (ViT), BERT
[90]	Introduces hyperbolic geometry for SER, combining hyperbolic and Euclidean layers for improved accuracy.	CMU-MOSI, CMU-MOSEI	Text, Audio, Video	Hyperbolic neural networks, Hyperbolic RNN, Hyperbolic Softmax

5.5. High complexity and computational cost

The processing and integration of numerous modalities in MMSA present a significant challenge due to high complexity and computational cost. It is difficult and costly to find long-term patterns in multimodal data. In order to solve these problems, Poria Peng et al. [91] developed a multimodal emotional data analysis framework that uses multiple kernel learning (MKL) at both the feature and decision levels to successfully integrate textual, auditory, and visual information. By integrating a deep convolutional neural network (CNN) and a support vector machine (SVM), it increases textual sentiment analysis accuracy while reducing computational overhead. To handle high processing demands, Sun Liu et al. [51] created a comparable modality completion-based MMSA (SMCMMSA) model. To add comparable modalities to missing data, it makes use of a Full Modal Samples Database (FMSD). SMCMMSA uses a multimodal feature fusion module to seamlessly merge text, audio, and video inputs, simplifying computations without sacrificing accuracy. The model’s decision-level fusion technique, which ensures accurate sentiment predictions even when data availability is low, demonstrates its capacity to stay effective in resource-constrained scenarios. For intra- and inter-modal feature learning, Yu Wei et al. [92] cre-

ated text and image bidirectional encoder representations from transformers (TIBERT). TIBERT combines unimodal and multimodal data to increase SA accuracy without excessive computer complexity. Its scalability and minimal resource needs are demonstrated by evaluations across numerous datasets, making it a flexible tool for a range of applications. Despite the outstanding performance of attention-based fusion and transformer models in MMSA, their self-attention mechanism develops quadratically with sequence length ($O(n^2)$) [40]. This can significantly increase inference latency and memory consumption, especially when processing high-frame-rate video streams or extended utterances [57]. Recent literature has provided lightweight solutions to lessen this effect. Tensor Fusion Networks (TFN) and Low-Rank Multimodal Fusion (LMF) [93] divide high-dimensional multimodal tensors into low-rank factors, significantly lowering the number of trainable parameters while preserving cross-modal interactions. Similarly, Multimodal Compact Bilinear (MCB) pooling [94] lowers the outer-product approximation of modalities to minimize computing overhead. When such lightweight approaches are added, models get outcomes that are close to state-of-the-art while still being suitable for real-time or resource-constrained modes. Table 7 provides an overview of methods for resolving complexity problems.

Table 7: Technique to handle high complexity and computational cost

Ref.	Contribution	Dataset	Modalities	DL Technique
[91]	Proposed a novel framework combining salient feature extraction from multiple modalities using MKL.	YouTube and IEMOCAP	Audio, Visual, Textual	SPG-GMKL, MK-ELM
[51]	Proposes SMCMSA, a model that completes uncertain missing modalities using similar modality completion and improves the quality of video and audio by fusing text representations.	CMU-MOSI, IEMOCAP	Text, Audio, Video	Modality completion, Transformer encoder, Decision-level fusion
[92]	Proposed a BERT-based text feature-guided multimodal sentiment analysis model (TIBERT) using feature-guided attention for better interaction and fusion of text and image data.	Weibo and MVSA-Single	Text and Image	Transformer-based BERT, ResNet, Multi-head attention mechanism

5.6. *Lack of explainability and interpretability*

It can be difficult to comprehend the logic behind the predictions made by current MMSA algorithms since they frequently lack explainability. MMSA models' limited explainability frequently prevents them from being used more broadly. Through the direct production of unimodal and bimodal sentiment decisions in addition to the overall sentiment, quantum-inspired frameworks, which use concepts like superposition and entanglement to simulate complicated interactions, provide a route to more interpretable models [95]. Complex emotions like sentiment and sarcasm in social media are still difficult for MMSA students to comprehend and interpret. Fuzzy logic and quantum neural networks are used by sophisticated models like the Quantum Fuzzy Neural Network (QFNN) to capture the ambiguity and complexity present in emotive and sardonic data, exhibiting improved interpretability and robustness [58]. Zhao et al.[96] proposed a multimodal aspect-based sentiment analysis (MASA) model to overcome this problem by utilizing layered interaction and adversarial training to identify the connections between text, images, and certain attributes. MASA improves interpretability and offers deeper insights into context-sensitive interactions by clearly demonstrating how each modality contributes to the final emotion prediction. In addition to helping with benchmarking, the release of the MASAD dataset encourages greater research into developing sentiment analysis systems that are more interpretable and context-aware. The Dual-channel sentiment analysis paradigm was presented by Wang Mengyi Wang [97] in order to make clear the roles that text, image, and audio data play in sentiment prediction. By using a three-way decision module to handle inter-modal conflicts, the framework ensures that its predictions are more transparent. Furthermore, the combination of adversarial training and soft Jaccard loss improves the differentiation of model outputs, leading to SA approaches that are more trustworthy and understandable. It illustrates the possibility of attaining higher interpretability and increased accuracy in MMSA. The approaches used for enhancing explainability and interoperability in MMSA are summarized in Table 8.

Table 8: Representative approaches with improved interpretability in MMSA.

Ref.	Contribution	Dataset	Modalities	DL Technique
[95]	Proposes a quantum-inspired multimodal fusion framework, using quantum superposition and entanglement to explicitly model intra- and inter-modal interactions for better interpretability.	CMU-MOSI, CMU-MOSEI	Text, Audio, Video	Quantum superposition, Quantum entanglement
[58]	Proposes QFNN using a hybrid classical-quantum neural network with fuzzy logic for enhanced sarcasm and sentiment feature extraction.	Mustard, Memotion	Text, Image, Audio	Quantum Neural Networks (QNN), Fuzzy Logic
[96]	Proposes MASAD, a large-scale multimodal dataset with both text and images, across seven domains and 57 aspects.	MASAD	Text and Images	Adversarial training, Multimodal interaction layers (text-image)
[97]	Introduces DCF, a dual-channel sentiment analysis framework, with visual-audio and multimodal models, and a three-way decision module to handle sentimental inconsistencies.	CMU-MOSI, CMU-MOSEI, UR-FUNNY, MUSTARD	Text, Audio, Video	Three-way decision, Adversarial training, Binary cross-entropy, Jaccard loss

5.7. Integration of sentiments and emotions

In MMSA, detecting changes in sentiment and emotional subtleties across multiple data modalities is still quite difficult. Combining sentiment with emotion can lead to problems with accuracy and consistency. To handle the intricacy of sentiment and emotion recognition in multimodal contexts, several creative techniques are required. Several strategies have been proposed by researchers to address these issues. A innovative way to resolving sentiment incongruity between text and visual data is offered by techniques like the Cross-modal Incongruity Perception Network (CiteNet), which specifically solves issues in emotion and sentiment recognition integration [98]. In order to ensure more accurate sentiment prediction, CiteNet’s cross-modal incongruity learning module analyzes underlying sentiment disparities across modalities and modifies the multimodal data fusion depending on these disparities. Techniques such as the Hierarchal Attention-BiLSTM (HALCB), which use task-specific fusion algorithms to divide sentiment analysis into binary and multi-classification modules, have demonstrated significant improvements in accuracy for both binary and multi-class [99]. A deep emotional arousal network (DEAN) is proposed by Feng et al. [56] to enhance the resilience and real-time adaptability of MMSA. To cut down on redundancy and improve prediction accuracy, DEAN employs a multimodal gating block that dynamically adjusts the weight of text, audio, and visual modalities

based on contextual relevance. Time-dependent interactions are incorporated into DEAN’s Transformer-based architecture, which successfully captures changes in emotional states over time. This makes it possible for the model to monitor and react to dynamic emotional shifts, which is essential for handling the emotional diversity that exists in the real world. Experiments on benchmark datasets show that DEAN is robust and can sustain excellent accuracy. Shao [100] proposes multimodal translation for sentiment analysis (MTSA) to overcome the difficulties in integrating sentiments and emotions. To bridge the gap between textual and non-textual aspects, MTSA uses BERT in place of traditional fusion techniques to convert visual and aural input into text-based representations. This translation procedure improves the model’s capacity to identify minor sentiment alterations and emotional nuances by allowing it to understand multimodal inputs coherently. Experiments on the CMU-MOSI and CMU-MOSEI datasets show MTSA integrates sentiment and emotion identification.

Table 9: Techniques to improve emotion and sentiment integration

Ref.	Contribution	Dataset	Modalities	DL Technique
[99]	Proposes the Hierarchical Attention-LSTM model based on Cognitive Brain (HALCB) to simulate brain processes with separate fusion strategies.	MOSI, MOSEI, YouTube	Text, Audio, Video	Attention-based BiLSTM, Multi-fusion strategies
[56]	A Deep Emotional Arousal Network (DEAN) that is capable of simulating the emotional coherence.	CMU-MOSI, CMU-MOSEI, IEMOCAP	Text, Image, Audio	Convolutional Neural Network (CNN), DEAN
[100]	Proposed Multimodal Translation for Sentiment Analysis (MTSA) framework.	CMU-MOSI, CMU-MOSEI	Text, Visual, Audio	Seq2Seq, Transformer, GRU
[98]	Proposed CiteNet: Cross-modal Incongruity Perception Network with extraction-estimation-fusion paradigm.	MVSA-Single, MVSA-Multiple	Text, Image	Cross-modal Alignment, Incongruity Learning, Integration

6. Statistical Analysis and Evaluation

Researchers have used several datasets and evaluation metrics for evaluating the performance of sentiment analysis models. Therefore, new researchers need to identify the most appropriate evaluation metric/s and dataset/s for evaluating their research. This section presents an overview and statistical analysis of the datasets and metrics used in the reviewed papers.

6.1. Datasets

This section discusses the multimodal sentiment analysis datasets used in the reviewed papers. Table 10 lists datasets, their brief discussion, and the references of the paper that used the dataset. It can be seen from the table that CMU-MOSI and CMU-MOSIE are the most popular datasets for multimodal sentiment analysis. These datasets include text, audio, and visual data, offering diverse modalities for sentiment and emotion recognition tasks. A brief overview of some popular MMSA datasets is presented in the subsections. Figure 7 presents the statistics showing the dataset and the number of times it has been used in the reviewed papers. If any particular dataset is used less than three times, then it is placed in the other category. This section will be helpful for researchers exploring the MMSA domain to find a suitable dataset for evaluating their model.

Though CMU-MOSI, CMU-MOSEI, and IEMOCAP are widely-used benchmarks, they do have biases, which would impact generalizability to heterogeneous realistic situations. These datasets have English content predominantly, and this limits potential usages in low-resource language and multilingual settings [85]. CMU-MOSI and MOSEI also have predominantly scripted or semi-scripted YouTube monologues, and little of this captures the spontaneity, code-switching, or noisy conditions typical in social media or realistic dialogue datasets [86]. Culture bias also seems latent, inasmuch as emotional output and prosody vary across languages and cultures, and models learned on these datasets would fare poorly in cross-cultural deployment. Future work in creating datasets must correct these weaknesses by having conversational, multilingual, and culturally representative samples to better strengthen models and fairness in models.

6.1.1. CMU-MOSI

CMU-MOSI is a multimodal sentiment analysis corpus consisting of 2,199 monologue utterances sampled from 93 YouTube movie review videos. The videos contain 89 English-speaking speakers from different nationalities. The corpus is rich with sentimental language, where each utterance is segmented according to long pauses and labeled by five human labelers on a continuous sentiment range of -3 (very negative) to +3 (very positive). It contains textual, acoustic (15 Hz), and visual (12.5 Hz) characteristics synchronized at the word level.

6.1.2. CMU-MOSEI

CMU-MOSEI is an expansion of the CMU-MOSI dataset with a larger, more varied set of multimodal sentiment data. It contains 23,453 labeled utterances organized into 3,228 monologue video segments, which are derived from more than 5,000 YouTube videos. The dataset features work from 1,000 distinct speakers on 250 different topics, which makes it one of the most diverse tools for sentiment and emotion analysis. Every sentence is labeled with a real-valued sentiment score between -3 (strongly negative) and +3 (strongly positive). CMU-MOSEI has textual, acoustic, and visual modalities, pre-processed with the CMU Multimodal SDK, word-aligned. With respect to CMU-MOSI, MOSEI provides much more linguistic, acoustic, and topical variability, facilitating stronger and more scalable multimodal understanding research.

6.1.3. IEMOCAP

The Interactive Emotional Dyadic Motion Capture (IEMOCAP) corpus is a large, acted multimodal emotion recognition corpus of the SAIL lab at USC. It includes about 12 hours of audiovisual recordings from 302 dyadic sessions with 10 professional actors participating in both scripted and improvisational scenarios aimed at provoking a large variety of emotional expressions.

It contains video, speech, facial motion capture, and text transcriptions as multimodal rich data for expressive human communication research. All the sessions are divided into utterances, and they have been annotated with both categorical emotion tags i.e., anger, happiness, sadness, neutral, excitement, frustration, fear, surprise, and disappointment—and dimensional scores such as valence, activation, and dominance by multiple annotators. To ensure research comparability and benchmarking, most papers concentrate on the main emotions: anger (1,083 samples), happiness (1,630), sadness (1,083), and neutral (1,683). Usually, training and evaluation are performed using only the data from the first eight speakers. The realism of IEMOCAP’s two-way interactions and richness of multimodal annotations make it possibly the most useful dataset for training and testing emotion recognition models in a multispeaker, multimodal environment.

6.1.4. TWITTER

Twitter15 and Twitter17 are two of the most popular benchmark datasets for target-specific multimodal sentiment analysis. Twitter15 has 3,179 training, 1,112 validation, and 1,037 test samples, and Twitter17 has 3,562 train-

ing, 1,176 validation, and 1,234 test samples. These have been the standard to use for model evaluation. For efficient learning of representations between modalities, multi-task learning methods have been utilized, improving performance in multimodal sentiment analysis tasks.

6.1.5. *MUStARD*

The MUStARD (Multimodal Sarcasm Detection) dataset is a well-curated set of 690 dialogue examples for multimodal sarcasm detection. The examples consist of a target utterance and context, including varied prior utterances. The dataset’s text, audio, and visual modalities make it a well-suited reference for sarcasm modeling in conversation. The corpus is taken from popular TV programs such as Friends, The Big Bang Theory, The Golden Girls, Buddies, and Sarcasmaholics Anonymous, and contains around 3.68 hours of data. All the samples are labeled by hand with binary sarcasm tags (sarcastic or non-sarcastic) with equal distribution of the two classes. This makes MUStARD an excellent resource for the analysis of subtle, context-dependent language like sarcasm in a multimodal environment.

6.1.6. *UR-FUNNY*

UR-FUNNY is a commonly applied multimodal humor detection corpus with 16,514 punchline excerpts from TED talks. Each excerpt is labeled with a binary value for the existence of humor: 0 for humorous and 1 for non-humorous. The corpus has a few modalities—text, audio, and visual signals—to enable solid analysis and classification of humorous content in oral speeches.

Table 10: Multimodal sentiment analysis datasets.

S. No	Dataset	Description	Reference
1	CMU-MOSI	Monologue videos with sentiment intensity (-3 to +3).	[56], [53], [57], [49], [55], [74], [83], [75], [84], [100], [59], [51], [97], [95], [99], [90]
2	CMU-MOSIE	Larger, diverse extension of CMU-MOSI with sentiment intensity.	[56], [53], [57], [49], [55], [74], [83], [75], [84], [100], [59], [97], [95], [99], [90]
3	IMOCAP	Dyadic conversations with emotion and sentiment annotations.	[56], [57], [91], [51]
4	MVSA	Social media posts with text-image sentiment labels.	[77]
5	VSA	Visual sentiment analysis dataset with captions.	[77]
6	Mvsa-Single	Single text-image pairs with sentiment labels.	[82], [92], [98]
7	MVSA-Multiple	Multiple text-image pairs with separate sentiment labels.	[40], [98]
8	Twitter15	Tweets with sentiment annotations and multimodal data.	[77], [76], [81]
9	Twitter17	Extension of Twitter15 with more annotated tweets.	[77], [76], [81]
10	Twitter19	Multimodal sarcasm detection with tweets.	[75]
11	T4SA	Multimodal tweets with sentiment labels.	[40]
12	Weibo	Chinese microblog posts with text and images.	[82], [92]
13	CH-SIMS	Chinese dataset with text and visual sentiment annotations.	[57], [59]
14	SEEmoji	Emoji usage in text paired with images for sentiment.	[89]
15	MUSARD	Sarcastic dialogues (text, audio, video) for sentiment.	[89], [58], [97]
16	TripAdvisor	Reviews with text and images for sentiment tasks.	[73]
17	MOUD	Spanish and English opinion utterances (audio, video, text).	[73]
18	Amigos	Videos annotated for sentiment, emotions, and signals.	[73]
19	MASAD	Sarcasm detection dataset with text and images.	[96]
20	Memotion	Meme dataset with text and images for sentiment/emotion.	[58]
21	UR-FUNNY	Humor detection with sentiment (text, visual, audio).	[97], [74], [83], [84]
22	Flickr	Images with captions for sentiment analysis.	[54], [86]
23	Getty Images	Professional images with descriptive captions.	[54], [86]
24	Twitter Multimodal Sarcasm Dataset	Multi-modal sarcasm detection using text, images, attributes.	[80]
25	MELD	Multi-party conversations labeled with emotions.	[83], [84]
26	NYT Gigaword Corpus	Event data from NYT Gigaword used for similarity evaluation.	[79]
27	eDAIC	Mental health disorder identification using multi-modal data.	[87]

6.2. Evaluation metrics

This section presents the most frequently used evaluation metrics in the reviewed papers. Table 12 shows that the most popular evaluation metrics are accuracy, F1-score, precision, recall, mean absolute error (MAE), and correlation. Table 12 also shows the performance of the reviewed MMSA models on the different evaluation metrics. It can be seen from the table that most of the MMSA models have used multiple metrics for evaluating the performance. It is notable that the use of more metrics for evaluating a

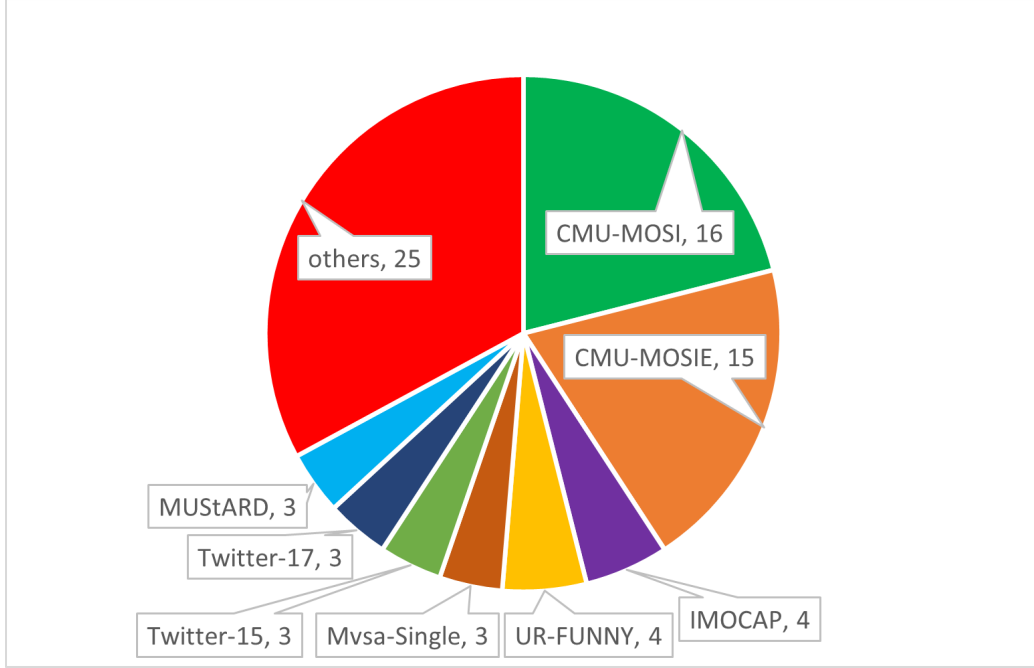


Figure 7: Multimodal sentiment analysis datasets.

model indicates that it is thoroughly analyzed. It also increases the understanding and reliability of the model. Figure 8 shows the usage percentage of different evaluation metrics in the reviewed papers. It can be observed from Figure 8 that accuracy highest usage percentage, whereas precision, recall, and correlation have the lowest usage percentage among the evaluation metrics. Accuracy remains as the leading reported metric. However, it obfuscates class imbalance and doesn't allow for subtle multimodal error types. F1-score offers a more balanced view for imbalanced datasets, and mean absolute error (MAE) can be utilized in sentiment intensity prediction. Increasing numbers of works report numerous measures to provide a general assessment, showing that one measure isn't sufficient in evaluating the performance of MMSA models.

Although performance comparisons are common, they often neglect crucial efficiency indicators like FLOPs, training time, and memory usage. The trade-offs between performance and efficiency are highlighted in Table 11, which collects reported and estimated values for several key models. For example, factorization techniques such as LMF [101] significantly decrease

Table 11: Efficiency indicators for selected MMSA models.

Model	Params (M)	Approx FLOPs	Training Time	Notes
TFN [102]	10–12M	High (tensor outer products)	–	Strong baseline but parameter explosion noted by authors.
LMF [101]	~3M	~10% of TFN	–	Explicitly designed for efficiency via low-rank factorization.
MulT [104]	30–40M	Very high (multi-stream transformers)	~1.5 days on 4 GPUs (MOSEI)	SOTA performance but compute-intensive.
MAG-BERT [105]	110M+ (BERT backbone)	Similar to BERT-base	Comparable to BERT fine-tuning	Efficiency depends largely on PLM size.
MISA [106]	50–60M	High	–	Extra representation heads increase cost relative to baselines.
SPT [103]	~25M	Reduced (sparse attention)	~30% faster than MulT	Explicit efficiency-oriented design.
AMTE [107]	40–45M	High	–	Very recent; efficiency not benchmarked widely.

the number of parameters compared to models like TFN [102]. On the other hand, sparse attention models like SPT [103] provide faster inference times. Conversely, models that heavily utilize transformers like MulT [104] and MAG-BERT [105] attain state-of-the-art accuracy, but they are computationally expensive.

6.3. Calibration metrics for reliable MMSA

Besides common measures like accuracy, F1-score, and Mean Absolute Error (MAE), model calibration becomes paramount to safety-critical MMSA applications. Calibration measures examine whether output probabilities match actual frequencies of outcomes. Expected Calibration Error (ECE)

and Maximum Calibration Error (MCE) bin predictions based on confidence and quantify errors from optimal calibration [Guo et al., 2017]. The Brier Score offers the correct scoring function, penalizing imprecise and poorly calibrated predictions. AUROC (Area Under the ROC Curve) and AUPRC (Area Under the Precision-Recall Curve) offer complementary information regarding class separability in imbalanced datasets [Snoek et al., 2019]. Reporting these, as well as accuracy-based measures, offers a richer estimation of trustworthiness in MMSA models, especially for healthcare, education, or affective computing applications for which estimates for confidence directly inform decisions.

Table 12: Evaluation Metrics used in MMSA

Ref.	Accuracy	F1-Score	Precision	Recall	MAE	Correlation
[56]	83.3	83.2	-	-	0.571	-
[77]	72.89	72.15	-	-	-	-
[53]	85.5	85.3	-	-	0.541	-
[40]	93-96	93-96	-	-	-	-
[82]	74.78	74.42	-	-	-	-
[57]	83.93	83.64	-	-	0.562	0.729
[89]	-	76.7	77.9	76.9	-	-
[78]	91.39	0.92	0.98	0.91	-	-
[54]	0.869	0.866	0.882	0.851	-	-
[91]	88.60	-	-	-	-	-
[45]	89.3	87.4	88.9	86.1	-	-
[73]	97.4	-	-	-	-	-
[92]	74.78	74.42	-	-	-	-
[49]	80	-	-	-	-	-
[96]	90.64	90.14	-	-	-	-
[55]	84.05	-	-	-	0.717	0.800
[86]	88.0	88.5	-	-	-	-
[100]	86.8	86.8	-	-	0.541	-
[58]	-	68.88	-	-	-	-
[59]	85.57	85.41	-	-	-	-
[51]	82.3	83.2	-	-	-	-
[97]	85.5	84.7	-	-	-	-
[95]	84.2	-	-	-	-	-
[99]	85.2	85.3	-	-	-	-
[90]	85.9	86.1	-	-	-	-
[74]	-	85	-	-	0.674	-
[83]	85.52	85.34	-	-	0.674	-
[80]	96.17	95.79	-	-	-	-
[75]	87.34	87.34	-	-	-	-
[79]	78.3	-	-	-	-	0.79
[76]	-	69.4	68.6	70.3	-	-
[84]	83.67	85.34	-	-	0.674	0.810
[81]	79.65	75.9	-	-	-	-
[98]	76.09	74.67	-	-	-	-
[87]	-	-	-	-	-	0.722

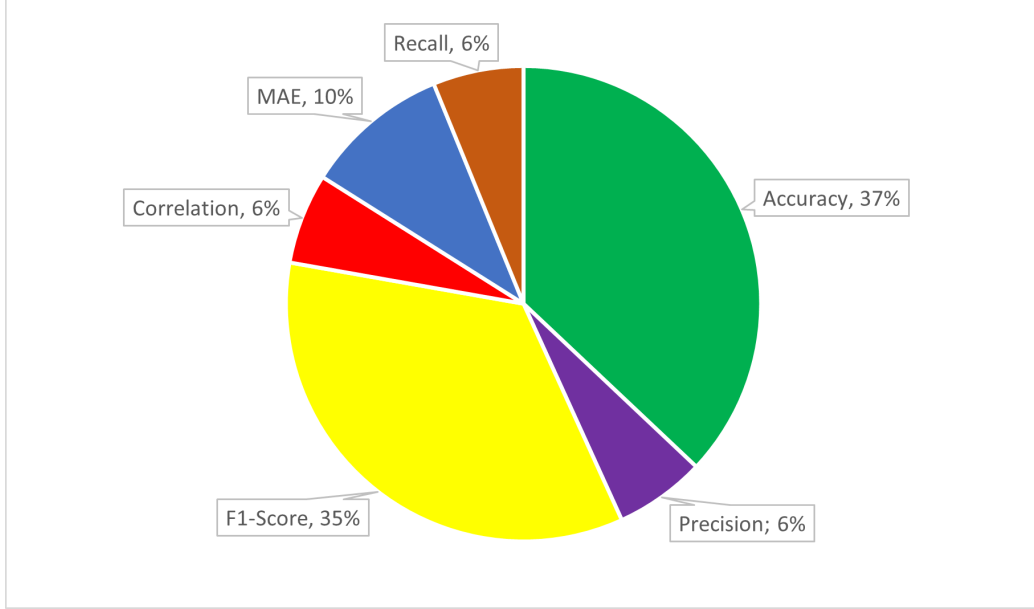


Figure 8: Evaluation metrics.

7. Future Research Perspectives

The incorporation of multimodal data in sentiment analysis models has improved their performance. However, there are several challenges and research perspectives that need the attention of researchers working in this domain. This section discusses some future research perspectives that can be explored by academicians and researchers working in sentiment analysis in general and multimodal sentiment analysis in particular. The prospective future research directions are briefly discussed in the subsequent subsections.

7.1. Improved fusion approaches

Traditional early and late fusion approaches often overlook the intricate interactions across different modalities. Future research should focus on developing advanced fusion techniques that effectively integrate multiple data modalities. To capture dynamic and context-aware interactions among modalities, emerging approaches include the use of graph neural networks, attention-based fusion, and tensor representations. These enhanced fusion techniques should lead to more reliable and comprehensible multimodal sentiment analysis by improving sentiment identification accuracy, particularly in situations with contradictory or unbalanced modalities.

7.2. Low resources languages

One of the most important research challenges in low-resource languages is sentiment analysis. The majority of current models are limited in their applicability in multilingual or underrepresented linguistic environments because they were trained on high-resource languages like English [108]. To close this gap, future research should concentrate on creating multilingual pre-trained models, transfer learning, and cross-lingual embeddings. Sentiment analysis performance for low-resource languages can be greatly enhanced by incorporating code-mixed data, utilizing limited annotated corpora through data augmentation, and adapting models via zero-shot or few-shot learning, encouraging inclusivity and worldwide applicability in multimodal systems.

7.3. Explainability

Ensuring model openness and interpretability becomes more crucial as multimodal sentiment analysis systems become more complicated. Although black-box models are frequently accurate, their inability to provide evidence for their predictions undermines user confidence and prevents adoption in delicate applications. In order to give human-understandable explanations for sentiment forecasts across modalities, future research should concentrate on combining explainable AI (XAI) methodologies [109]. Techniques like concept-based explanations, modality attribution, and attention visualization can provide information on how particular textual, auditory, or visual cues affect the result. Improving explainability helps troubleshoot and improve model behavior in addition to building user trust [110].

7.4. Intra-modality dynamics

Capturing the intra-modality fluctuations inside a single modality is a major challenge in multimodal sentiment analysis. In spoken language, these dynamics can be especially challenging to understand [102]. For instance, depending on the speaker’s tone and facial expression, the statement "I can’t believe this happened" can suggest astonishment, exhilaration, or dissatisfaction. The speaker may convey enthusiasm if they say it with wide eyes and a happy tone, or disappointment if they say it with a flat tone and a sigh. In visual and audio modalities, where minute changes occur in both spatial and temporal settings, intra-modality dynamics are also very important.

7.5. Inter-modality dynamics

Modeling intra-modal dynamics while accurately capturing inter-modal interactions is the second major problem in multimodal sentiment analysis. Understanding how several modalities—such as text, auditory, and visual cues—interact and affect the overall sentiment interpretation is the task in question. In spoken language, where temporal and prosodic features frequently diverge from structured writing forms, this becomes especially complicated. A spoken statement such as "Um... I'm not entirely sure... maybe it was fine... or not..." is an example of hesitation, self-correction, and doubt, all of which make linguistic analysis more difficult. It is challenging to recover coherent sentiment representations from the textual modality alone because of these disfluent patterns that are common in spontaneous speech. Accurately modeling such inter-modality dependencies is essential for reliable sentiment inference in multimodal contexts.

7.6. Personality or mood detection

It is possible to deduce a person's underlying personality traits or present mood states by using emotion recognition from textual data. Social media sites and other user-focused apps can employ this emotive analysis to provide tailored, context-aware recommendations. Systems can improve user engagement through more targeted decision support, better human-computer interaction, and adaptable content delivery by integrating personality or mood analysis.

7.7. Sentiment-cause detection

The accuracy of sentiment categorization in spoken and textual data can be greatly improved by determining the underlying reason or trigger of a feeling. By comprehending the particular context or event causing the sentiment, causal sentiment analysis enables models to go beyond generic polarity detection. For example, recognizing that the positive feeling in the statement "I'm thrilled—I just got promoted at work" is associated with a job promotion allows for a more accurate and contextually relevant interpretation. Sentiment-aware systems are more effective and easier to understand when such causal reasoning is incorporated.

7.8. Multiple sentiment detection

Finding the dominating sentiment in the text is the main goal of the majority of sentiment detection systems. Sentences that convey more than

one sentiment are frequently rejected or categorized according to the first sentiment that is identified. However, the system should be able to identify both positive and negative thoughts while taking into account their temporal and contextual linkages in situations like "I was excited about the event, but now I feel a bit disappointed" or "I love this product, but the delivery took too long." In order to improve the system's capacity to handle complicated, mixed-sentiment expressions, this calls for sophisticated models that can comprehend sentiment transitions and simultaneous sentiments.

7.9. Complexity

Scalability, interpretability, and deployment efficiency are all hampered by the growing complexity of multimodal sentiment analysis models. Large annotated datasets and substantial computational resources are frequently needed for deep architectures that include many modalities, which may restrict their usefulness in real-world or resource-constrained settings. Future studies should focus on creating efficient, lightweight models that strike a compromise between computational economy and performance. Modular architectures, knowledge distillation, and model pruning are among strategies that can assist cut complexity without sacrificing accuracy. Building scalable, accessible, and long-lasting sentiment analysis systems requires addressing complexity.

7.10. Dataset

The existing multimodal datasets are good for sentiment analysis, but most of them are only in specific domains or types of content. Future research should consider more diverse and large-scale datasets, especially those that have a wider variety of multimodal inputs, such as user-generated content from several social media platforms. Second, datasets with more nuanced emotional expressions, such as sarcasm and mixed emotions, might have better generalization capabilities of the models. It will also be very useful if future datasets include less frequently considered modalities, such as video or audio, to better judge how the model can be generalized to more complex multimodal fusion tasks.

7.11. Cross-cultural and cross-lingual SA

Sentiment is very culture-sensitive and context-sensitive, and cross-cultural and cross-lingual MMSA is therefore an uphill research field. As an example, irony, sarcasm, or emotional strength expressions differ across cultures and

would not likely be modeled by systems whose training datasets are overwhelmingly English datasets like CMU-MOSI or MOSEI. This leaves us at an impasse regarding generalizability and fairness in sentiment estimation. Additionally, few low-resource languages have multimodal corpora, and this restricts us to having imbalanced datasets to train and test models upon. Recent works have sought to overcome these challenges through multilingual pre-trained language models (like multilingual BERT, XLM-R) and multimodal fusion. Others seek to apply transfer learning across cultures and languages, or cultural knowledge graphs to supply contextual grounding. While these works have made some headway, strong cross-cultural MMSA remains an open research avenue, needing more extensive and varied datasets, new transfer paradigms, and deeper cultural semantic exploration.

7.12. Real-world applications and robustness

Deploying multimodal sentiment analysis systems in real-world applications requires robust performance under diverse and unpredictable conditions. Variations in speech accents, lighting, background noise, and informal or code-mixed language can degrade model accuracy. Future research should emphasize building models that generalize well across domains, user demographics, and noisy environments. Robustness can be improved by using adversarial training, domain adaptability, and ongoing learning. Furthermore, to provide practical and dependable deployment in dynamic, real-world situations, system development must be in line with application-specific needs, such as real-time processing for customer service or adaptability in social robots.

7.13. Transfer learning and fine-tuning

In multimodal data analysis, transfer learning and fine-tuning have become essential tactics to overcome data scarcity and enhance model generalization [111]. Effective adaptation to certain target domains with less annotated data is made possible by utilizing pre-trained models on large-scale datasets. More efficient fine-tuning methods that take into account cross-modal interdependence and modality-specific subtleties should be investigated in future studies. Furthermore, maintaining performance in the face of changing real-world data distributions will need the development of adaptive transfer learning frameworks that are capable of ongoing learning and domain adaptation. This will improve the flexibility and scalability of sentiment analysis systems.

8. Conclusion

MMSA is one of the most researched areas of the last decade that utilizes multiple data modalities like text, audio, and visual to enhance sentiment analysis. This paper extensively reviewed MMSA, discussing data modalities, fusion approaches, and their associated strengths and weaknesses. In addition, it identifies critical challenges, such as modality heterogeneity, inconsistent data quality, text-centric models, synchronization, fusion complexity, and lack of explainability and interpretability and reviews a number of techniques proposed by researchers to address these issues. It also presented a general framework for MMSA, discussing its different steps in detail. Moreover, it identified benchmark datasets for MMSA, evaluation metrics, and presented their statistical analysis. Furthermore, it proposed several future research perspectives like improved fusion approaches, explainability, inter-modality dynamics, intra-modality dynamics, development of better datasets and more to enhance the MMSA. It is believed that this paper will be useful for researchers exploring sentiment analysis and multimodal information retrieval in general and MMSA in particular.

References

- [1] K. Zhang, J. Li, Z. Li, J. Zhang, Composed multi-modal retrieval: A survey of approaches and applications, arXiv preprint arXiv:2503.01334 (2025).
- [2] K. Anwar, M. Wasid, A. Zafar, M. Ganaie, A. Iqbal, Soft computing techniques in multi-criteria recommender systems: A comprehensive review, *Applied Soft Computing* (2024) 112579.
- [3] B. Liang, H. Su, L. Gui, E. Cambria, R. Xu, Aspect-based sentiment analysis via affective knowledge enhanced graph convolutional networks, *Knowledge-Based Systems* 235 (2022) 107643.
- [4] A. Yadollahi, A. G. Shahraki, O. R. Zaiane, Current state of text sentiment analysis from opinion to emotion mining, *ACM Computing Surveys (CSUR)* 50 (2) (2017) 1–33.
- [5] K. Anwar, A. Garg, S. Ahmad, M. Wasid, Machine learning models for sentiment analysis: State of the art and applications, in: 2024 15th In-

ternational Conference on Computing Communication and Networking Technologies (ICCCNT), 2024, pp. 1–6.

- [6] K. Du, Y. Chen, Y. Zhang, X. Li, Financial sentiment analysis: Techniques and applications, *ACM Computing Surveys* 56 (9) (2024) 1–42.
- [7] J. Cen, Q. Li, J. Luo, M. Zhang, W. Ma, Masanet: Multi-aspect semantic auxiliary network for visual sentiment analysis, *IEEE Transactions on Affective Computing* 15 (3) (2024) 1439–1450.
- [8] S. K. Baberwal, N. A. Shelke, K. Anwar, Systematic review of recent advances in multimodal sentiment analysis, *Discover Computing* 28 (1) (2025) 270.
- [9] J. Du, J. Jin, J. Zhuang, C. Zhang, Hierarchical graph contrastive learning of local and global presentation for multimodal sentiment analysis, *Scientific Reports* 14 (1) (2024) 5335.
- [10] S. Poria, D. Hazarika, N. Majumder, R. Mihalcea, Beneath the tip of the iceberg: Current challenges and new directions in sentiment analysis research, *IEEE Transactions on Affective Computing* 14 (1) (2023) 108–132.
- [11] M. P. Fortin, B. Chaib-Draa, Multimodal sentiment analysis: A multitask learning approach, in: *ICPRAM*, 2019, pp. 368–376.
- [12] S. Liu, Z. Luo, W. Fu, Fcdnet: Fuzzy cognition-based dynamic fusion network for multimodal sentiment analysis, *IEEE Transactions on Fuzzy Systems* 33 (1) (2025) 3–14.
- [13] L. Stappen, L. Schumann, B. Sertolli, A. Baird, B. Weigell, E. Cambria, B. W. Schuller, Muse-toolbox: The multimodal sentiment analysis continuous annotation fusion and discrete class transformation toolbox, in: *Proceedings of the 2nd on Multimodal Sentiment Analysis Challenge*, 2021, pp. 75–82.
- [14] H. Abburi, R. Prasath, M. Shrivastava, S. V. Gangashetty, Multimodal sentiment analysis using deep neural networks, in: *Mining Intelligence and Knowledge Exploration: 4th International Conference, MIKE 2016, Mexico City, Mexico, November 13-19, 2016, Revised Selected Papers 4*, Springer, 2017, pp. 58–65.

- [15] S. Lai, X. Hu, H. Xu, Z. Ren, Z. Liu, Multimodal sentiment analysis: A survey, *Displays* 80 (2023) 102563.
- [16] R. Das, T. D. Singh, Multimodal sentiment analysis: a survey of methods, trends, and challenges, *ACM Computing Surveys* 55 (13s) (2023) 1–38.
- [17] L. Zhu, Z. Zhu, C. Zhang, Y. Xu, X. Kong, Multimodal sentiment analysis based on fusion methods: A survey, *Information Fusion* 95 (2023) 306–325.
- [18] A. Gandhi, K. Adhvaryu, S. Poria, E. Cambria, A. Hussain, Multimodal sentiment analysis: A systematic review of history, datasets, multimodal fusion methods, applications, challenges and future directions, *Information Fusion* 91 (2023) 424–444.
- [19] S. A. Abdu, A. H. Yousef, A. Salem, Multimodal video sentiment analysis using deep learning approaches, a survey, *Information Fusion* 76 (2021) 204–226.
- [20] A. Pandey, D. K. Vishwakarma, Progress, achievements, and challenges in multimodal sentiment analysis using deep learning: A survey, *Applied Soft Computing* 152 (2024) 111206.
- [21] T. Zhao, L.-a. Meng, D. Song, Multimodal aspect-based sentiment analysis: A survey of tasks, methods, challenges and future directions, *Information Fusion* 112 (2024) 102552.
- [22] S. Liu, P. Gao, Y. Li, W. Fu, W. Ding, Multi-modal fusion network with complementarity and importance for emotion recognition, *Information Sciences* 619 (2023) 679–694.
- [23] L. Xiao, X. Wu, W. Wu, J. Yang, L. He, Multi-channel attentive graph convolutional network with sentiment fusion for multimodal sentiment analysis, in: *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2022, pp. 4578–4582.
- [24] H. Lin, P. Zhang, J. Ling, Z. Yang, L. K. Lee, W. Liu, Ps-mixer: A polar-vector and strength-vector mixer model for multimodal sentiment analysis, *Information Processing & Management* 60 (2) (2023) 103229.

- [25] K. Santosh, R. Rizk, S. K. Bajracharya, Understanding Data—Modalities and Preprocessing, Springer Nature Singapore, Singapore, 2024, pp. 13–24.
- [26] M. Yasir, A. Zafar, K. Anwar, S. Parvez, Sadwc: sentiment analysis approach to detect depression using whatsapp chat data, International Journal of System Assurance Engineering and Management (2025) 1–15.
- [27] M. Wankhade, A. C. S. Rao, C. Kulkarni, A survey on sentiment analysis methods, applications, and challenges, Artificial Intelligence Review 55 (7) (2022) 5731–5780.
- [28] H. F. T. Al-Saadawi, B. Das, R. Das, A systematic review of trimodal affective computing approaches: Text, audio, and visual integration in emotion recognition and sentiment analysis, Expert Systems with Applications (2024) 124852.
- [29] F. Lin, S. Liu, C. Zhang, J. Fan, Z. Wu, Stylebert: Text-audio sentiment analysis with bi-directional style enhancement, Information systems 114 (2023) 102147.
- [30] A. Ortis, G. M. Farinella, G. Torrisi, S. Battiato, Exploiting objective text description of images for visual sentiment analysis, Multimedia Tools and Applications 80 (15) (2021) 22323–22346.
- [31] D. Chen, C. Yu, D. Xu, X. Wang, Joint multimodal sentiment analysis based on information relevance, Information Processing & Management 60 (2) (2023) 103193.
- [32] F. Zhao, C. Zhang, B. Geng, Deep multimodal data fusion, ACM Computing Surveys 56 (9) (2024) 1–36.
- [33] K. Zhao, M. Zheng, Q. Li, J. Liu, Multimodal sentiment analysis-a comprehensive survey from a fusion methods perspective, IEEE Access PP (2025) 1–1.
- [34] Z. Liu, B. Zhou, L. Meng, G. Huang, Multimodal sentiment analysis using bigru and attention-based hybrid fusion strategy, Intelligent Automation & Soft Computing 37 (2) (2023) 1963–1981.

- [35] L. Stappen, A. Baird, L. Schumann, B. Schuller, The multimodal sentiment analysis in car reviews (muse-car) dataset: Collection, insights and improvements, *IEEE Transactions on Affective Computing* 14 (2) (2021) 1334–1350.
- [36] X. Yan, H. Xue, S. Jiang, Z. Liu, Multimodal sentiment analysis using multi-tensor fusion network with cross-modal modeling, *Applied Artificial Intelligence* 36 (1) (2022) 2000688.
- [37] Z. Quan, T. Sun, M. Su, J. Wei, Multimodal sentiment analysis based on cross-modal attention and gated cyclic hierarchical fusion networks, *Computational Intelligence and Neuroscience* 2022 (2022) 4767437.
- [38] G. Paraskevopoulos, E. Georgiou, A. Potamianos, Mmlatch: Bottom-up top-down fusion for multimodal sentiment analysis (2022).
- [39] A. Roitberg, K. Peng, Z. Marinov, C. Seibold, D. Schneider, R. Stiefelhagen, A comparative analysis of decision-level fusion for multimodal driver behaviour understanding (2022).
- [40] A. Ghorbanali, M. K. Sohrabi, F. Yaghmaee, Ensemble transfer learning-based multimodal sentiment analysis using weighted convolutional neural networks, *Information Processing and Management* 59 (2022) 102929.
- [41] E. M. G. Younis, S. M. Zaki, E. Kanjo, E. H. Houssein, Evaluating ensemble learning methods for multi-modal emotion recognition using sensor data fusion, *Sensors* 22 (15) (2022) 5611.
- [42] S. Li, H. Tang, Multimodal alignment and fusion: A survey, *arXiv preprint* (2024).
- [43] S. Shankar, L. Thompson, M. Fiterau, Progressive fusion for multimodal integration (2022).
- [44] X. Hu, M. Yamamura, Global local fusion neural network for multimodal sentiment analysis, *Applied Sciences* 12 (17) (2022) 8453.
- [45] R. Das, T. D. Singh, A multi-stage multimodal framework for sentiment analysis of assamese in low resource setting, *Expert Systems With Applications* 204 (2022) 117575.

- [46] M. Xu, F. Liang, X. Su, C. Fang, Cmjrt: Cross-modal joint representation transformer for multimodal sentiment analysis, *IEEE Access* 10 (2022) 131671–131679.
- [47] M. Jiang, S. Ji, Cross-modality gated attention fusion for multimodal sentiment analysis (2022).
- [48] Z. Quan, T. Sun, M. Su, J. Wei, Multimodal sentiment analysis based on cross-modal attention and gated cyclic hierarchical fusion networks, *Computational Intelligence and Neuroscience* 2022 (2022) 4767437.
- [49] N. Majumder, D. Hazarika, A. Gelbukh, E. Cambria, S. Poria, Multimodal sentiment analysis using hierarchical fusion with context modeling, *Knowledge-Based Systems* 164 (2018) 164–178.
- [50] R. Lin, H. Hu, Dynamically shifting multimodal representations via hybrid-modal attention for multimodal sentiment analysis, *IEEE Transactions on Multimedia* (2023).
- [51] Y. Sun, Z. Liu, Q. Z. Sheng, D. Chu, J. Yu, H. Sun, Similar modality completion-based multimodal sentiment analysis under uncertain missing modalities, *Information Fusion* 110 (2024) 102454.
- [52] Z. Liu, B. Zhou, L. Meng, G. Huang, Multimodal sentiment analysis using bigru and attention-based hybrid fusion strategy, *Intelligent Automation & Soft Computing* 37 (2) (2023) 1963–1981.
- [53] B. Yang, B. Shao, L. Wu, X. Lin, Multimodal sentiment analysis with unidirectional modality translation, *Neurocomputing* 467 (2022) 130–137.
- [54] F. Huang, X. Zhang, Z. Zhao, J. Xu, Z. Li, Image–text sentiment analysis via deep multimodal attentive fusion, *Knowledge-Based Systems* 165 (2019) 230–241.
- [55] D. Wang, X. Guo, Y. Tian, J. Liu, L. He, X. Luo, Tetfn: A text enhanced transformer fusion network for multimodal sentiment analysis, *Pattern Recognition* 138 (2022) 109259.
- [56] F. Zhang, X.-C. Li, C. P. Lim, Q. Hua, C.-R. Dong, J.-H. Zhai, Deep emotional arousal network for multimodal sentiment analysis and emotion recognition, *Information Fusion* 88 (2022) 296–304.

- [57] T. Wu, J. Peng, W. Zhang, H. Zhang, S. Tan, F. Yi, C. Ma, Y. Huang, Videosentiment analysis with bimodal information-augmented multi-head attention, *Knowledge-Based Systems* 235 (2022) 107676.
- [58] P. Tiwari, L. Zhang, Z. Qu, G. Muhammad, Quantum fuzzy neural network for multimodal sentiment and sarcasm detection, *Information Fusion* 88 (2023) 102085.
- [59] J. Hou, N. Omar, S. Tiun, S. Saad, Q. He, Tchfn: Multimodal sentiment analysis based on text-centric hierarchical fusion network, *Knowledge-Based Systems* 300 (2024) 112220.
- [60] K. Zhao, M. Zheng, Q. Li, J. Liu, Multimodal sentiment analysis-a comprehensive survey from a fusion methods perspective, *IEEE Access PP* (2025) 1–1.
- [61] Y. Cai, X. Li, Y. Zhang, J. Li, F. Zhu, L. Rao, Multimodal sentiment analysis based on multi-layer feature fusion and multi-task learning, *Scientific Reports* 15 (1) (2025) 2126.
- [62] J. Fu, Y. Fu, H. Xue, et al., Tmfn: A text-based multimodal fusion network with multi-scale feature extraction and unsupervised contrastive learning for multimodal sentiment analysis, *Complex Intelligent Systems* 11 (2025) 133.
- [63] H. Lee, S. Suniljit, Y. S. Ong, Dynamic multimodal sentiment analysis: Leveraging cross-modal attention for enabled classification (2025).
- [64] C. Lu, X. Fu, Sentdep: Pioneering fusion-centric multimodal sentiment analysis for unprecedented performance and insights, *IEEE Access PP* (2024) 1–1.
- [65] M. T. Ribeiro, S. Singh, C. Guestrin, " why should i trust you?" explaining the predictions of any classifier, in: *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 2016, pp. 1135–1144.
- [66] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, *Advances in neural information processing systems* 30 (2017).

- [67] I. K. S. Al-Tameemi, M.-R. Feizi-Derakhshi, S. Pashazadeh, M. Asadpour, Interpretable multimodal sentiment classification using deep multi-view attentive network of image and text data, *IEEE Access* 11 (2023).
- [68] C.-D. Nguyen, T. Nguyen, D. Vu, A. Luu, Improving multimodal sentiment analysis: Supervised angular margin-based contrastive learning for enhanced fusion representation, in: H. Bouamor, J. Pino, K. Bali (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2023*, Association for Computational Linguistics, Singapore, 2023, pp. 14714–14724.
- [69] Y. Yu, M. Zhao, S.-a. Qi, F. Sun, B. Wang, W. Guo, X. Wang, L. Yang, D. Niu, Conki: Contrastive knowledge injection for multimodal sentiment analysis, in: A. Rogers, J. Boyd-Graber, N. Okazaki (Eds.), *Findings of the Association for Computational Linguistics: ACL 2023*, Association for Computational Linguistics, Toronto, Canada, 2023, pp. 13610–13624.
- [70] F. T. J. Faria, L. H. Baniata, M. H. Baniata, M. A. Khair, A. I. Bani Ata, C. Bunternghit, S. Kang, Sentimentformer: A transformer-based multimodal fusion framework for enhanced sentiment analysis of memes in under-resourced bangla language, *Electronics* 14 (4) (2025) 799.
- [71] F. Zhao, C. Zhang, B. Geng, Modality adaptation in multimodal data, *Expert Systems with Applications* 176 (2021) 114803.
- [72] X. Zhao, Y. Chen, W. Li, L. Gao, B. Tang, Mag+: An extended multimodal adaptation gate for multimodal sentiment analysis, in: *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2022, pp. 4753–4757.
- [73] I. Chaturvedi, R. Satapathy, S. Cavallari, E. Cambria, Fuzzy common-sense reasoning for multimodal sentiment analysis, *Pattern Recognition Letters* 125 (2019) 264–270.
- [74] C. Huang, J. Chen, Q. Huang, S. Wang, Y. Tu, X. Huang, Atcaf: Attention-based causality-aware fusion network for multimodal sentiment analysis, *Information Fusion* 114 (2025) 102725.

- [75] C. Zhu, M. Chen, S. Zhang, C. Sun, H. Liang, Y. Liu, J. Chen, Skeafn: Sentiment knowledge enhanced attention fusion network for multimodal sentiment analysis, *Information Fusion* 100 (2023) 101958.
- [76] L. Xiao, X. Wu, J. Xu, W. Li, C. Jin, L. He, Atlantis: Aesthetic-oriented multiple granularities fusion network for joint multimodal aspect-based sentiment analysis, *Information Fusion* 100 (2024) 102304.
- [77] J. Yea, J. Zhou, J. Tian, R. Wang, J. Zhou, T. Gui, Q. Zhang, X. Huang, Sentiment-aware multimodal pre-training for multimodal sentiment analysis, *Knowledge-Based Systems* 258 (2022) 110021.
- [78] K. Dashtipour, M. Gogate, E. Cambri, A. Hussain, A novel context-aware multimodal framework for persian sentiment analysis, *Neurocomputing* 449 (2021) 234–246.
- [79] Y. Zhou, X. Li, Sentimental contrastive learning for event representation, *Natural Language Processing* 100 (2023) 100031.
- [80] H. Liu, R. Wei, G. Tu, J. Lin, C. Liu, D. Jiang, Sarcasm driven by sentiment: A sentiment-aware hierarchical fusion network for multimodal sarcasm detection, *Information Fusion* 114 (2025) 102725.
- [81] Y. Weng, L. Chen, S. Wang, X. Ye, X. Liu, Z. Liu, Chaomurilige, Miecfc: Multi-faceted information extraction and cross-mixture fusion for multimodal aspect-based sentiment analysis, *Heliyon* 10 (2024) e32967.
- [82] B. Yu, J. Wei, B. Yu, X. Cai, K. Wang, H. Sun, L. Bu, X. Chen, Feature-guided multimodal sentiment analysis towards industry 4.0, *Computers and Electrical Engineering* 102 (2022) 107961.
- [83] Y. Zhou, X. Liang, H. Chen, Y. Zhao, X. Chen, L. Yu, Triple disentangled representation learning for multimodal affective analysis, *Information Fusion* 114 (2025) 102725.
- [84] Y. Zhou, X. Liang, H. Chen, Y. Zhao, X. Chen, L. Yu, Triple disentangled representation learning for multimodal affective analysis, *Information Fusion* 100 (2024) 102663.
- [85] R. Lin, H. Hu, Missmodal: Increasing robustness to missing modality in multimodal sentiment analysis, *Transactions of the Association for Computational Linguistics* 11 (2023) 1686–1702.

- [86] Y. Liu, Z. Li, K. Zhou, L. Zhang, L. Li, P. Tian, S. Shen, Scanning, attention, and reasoning multimodal content for sentiment analysis, *Knowledge-Based Systems* 268 (2023) 110467.
- [87] M. Dia, G. Khodabandelou, A. Othmani, Paying attention to uncertainty: A stochastic multimodal transformers for post-traumatic stress disorder detection using video, *Computer Methods and Programs in Biomedicine* 208 (2024) 108439.
- [88] R. Huan, G. Zhong, P. Chen, R. Liang, Unimf: A unified multimodal framework for multimodal sentiment analysis in missing modalities and unaligned multimodal sequences, *IEEE Transactions on Multimedia* (2023).
- [89] D. S. Chauhan, G. V. Singh, A. Arora, A. Ekbal, P. Bhattacharyya, An emoji-aware multitask framework for multimodal sarcasm detection, *Knowledge-Based Systems* 257 (2022) 109924.
- [90] K. A. Arano, C. Orsenigo, M. Soto, C. Vercellis, Multimodal sentiment and emotion recognition in hyperbolic space, *Expert Systems with Applications* 180 (2021) 115507.
- [91] S. Poria, H. Peng, A. Hussain, N. Howard, E. Cambria, Ensemble application of convolutional neural networks and multiple kernel learning for multimodal sentiment analysis, *Neurocomputing* 261 (2017) 217–230.
- [92] B. Yu, J. Wei, B. Yu, X. Cai, K. Wang, H. Sun, L. Bu, X. Chen, Feature-guided multimodal sentiment analysis towards industry 4.0, *Computers and Electrical Engineering* 102 (2022) 107961.
- [93] X. Miao, X. Zhang, H. Zhang, Low-rank tensor fusion and self-supervised multi-task multimodal sentiment analysis, *Multimedia Tools and Applications* 83 (23) (2024) 63291–63308.
- [94] Y. Zhang, C. Cheng, S. Wang, T. Xia, Emotion recognition using heterogeneous convolutional neural networks combined with multimodal factorized bilinear pooling, *Biomedical Signal Processing and Control* 77 (2022) 103877.

- [95] Q. Li, D. Gkoumas, C. Lioma, M. Melucci, Quantum-inspired multimodal fusion for video sentiment analysis, *Information Fusion* 110 (2024) 102454.
- [96] J. Zhou, J. Zhao, J. X. Huang, Q. V. Hu, L. He, Masad: A large-scale dataset for multimodal aspect-based sentiment analysis, *Neurocomputing* 463 (2021) 13–23.
- [97] X. Wang, M. Wang, H. Cui, Y. Zhang, A dual-channel multimodal sentiment analysis framework based on three-way decision, *Engineering Applications of Artificial Intelligence* 137 (2024) 109174.
- [98] J. Wang, Y. Yang, K. Liu, Z. Xie, F. Zhang, T. Li, Citenet: Cross-modal incongruity perception network for multimodal sentiment prediction, *Knowledge-Based Systems* 100 (2024) 111848.
- [99] Y. Li, K. Zhang, J. Wang, X. Gao, A cognitive brain model for multimodal sentiment analysis based on attention neural networks, *Neurocomputing* 430 (2021) 159–173.
- [100] B. Yang, B. Shao, L. Wu, X. Lin, Multimodal sentiment analysis with unidirectional modality translation, *Neurocomputing* 467 (2022) 130–137.
- [101] Z. Liu, Y. Shen, V. E. Lakshminarasimhan, P. P. Liang, A. Zadeh, L.-P. Morency, Efficient low-rank multimodal fusion with modality-specific factors, in: *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018, pp. 2247–2256.
- [102] A. Zadeh, M. Chen, S. Poria, E. Cambria, L.-P. Morency, Tensor fusion network for multimodal sentiment analysis, *arXiv preprint arXiv:1707.07250* (2017).
- [103] M. Lan, Z. Sun, R. He, Sparse phased transformer for multimodal sentiment analysis, in: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021, pp. 104–115.
- [104] Y.-H. H. Tsai, S. Bai, P. P. Liang, J. Z. Kolter, L.-P. Morency, R. Salakhutdinov, Multimodal transformer for unaligned multimodal

- language sequences, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL), 2019, pp. 6558–6569.
- [105] W. Rahman, M. K. Hasan, S. Lee, A. Zadeh, L.-P. Morency, Integrating multimodal information in large pretrained transformers with modality adaptation gate, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 2359–2375.
 - [106] D. Hazarika, R. Zimmermann, S. Poria, Misa: Modality-invariant and -specific representations for multimodal sentiment analysis, in: Proceedings of the 28th ACM International Conference on Multimedia (ACM MM), 2020, pp. 1122–1131.
 - [107] A. Tuerhong, Y. Zhang, L. Gao, Adaptive multimodal transformer based on exchanging for sentiment analysis, Scientific Reports 15 (1) (2025) 1–14.
 - [108] V. Vetrivel, Sentiment analysis on a low-resource language dataset using multimodal representation learning and cross-lingual transfer learning, Applied Soft Computing 157 (2024) 111553.
 - [109] E. Hashmi, S. Yildirim Yayilgan, A robust hybrid approach with product context-aware learning and explainable ai for sentiment analysis in amazon user reviews, Electronic Commerce Research (2024) 1–33.
 - [110] S. Sun, et al., A review of multimodal explainable artificial intelligence: Past, present and future, arXiv preprint arXiv:2412.14056 (2024).
 - [111] N. Sharma, B. Verma, Recent advances in transfer learning for natural language processing (nlp), in: A Handbook of Computational Linguistics: Artificial Intelligence in Natural Language Processing, Springer, 2024, pp. 228–254.

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